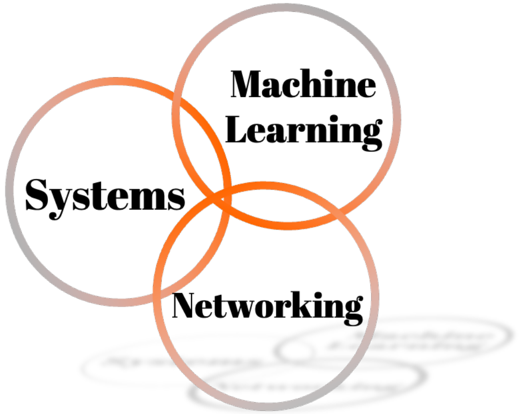
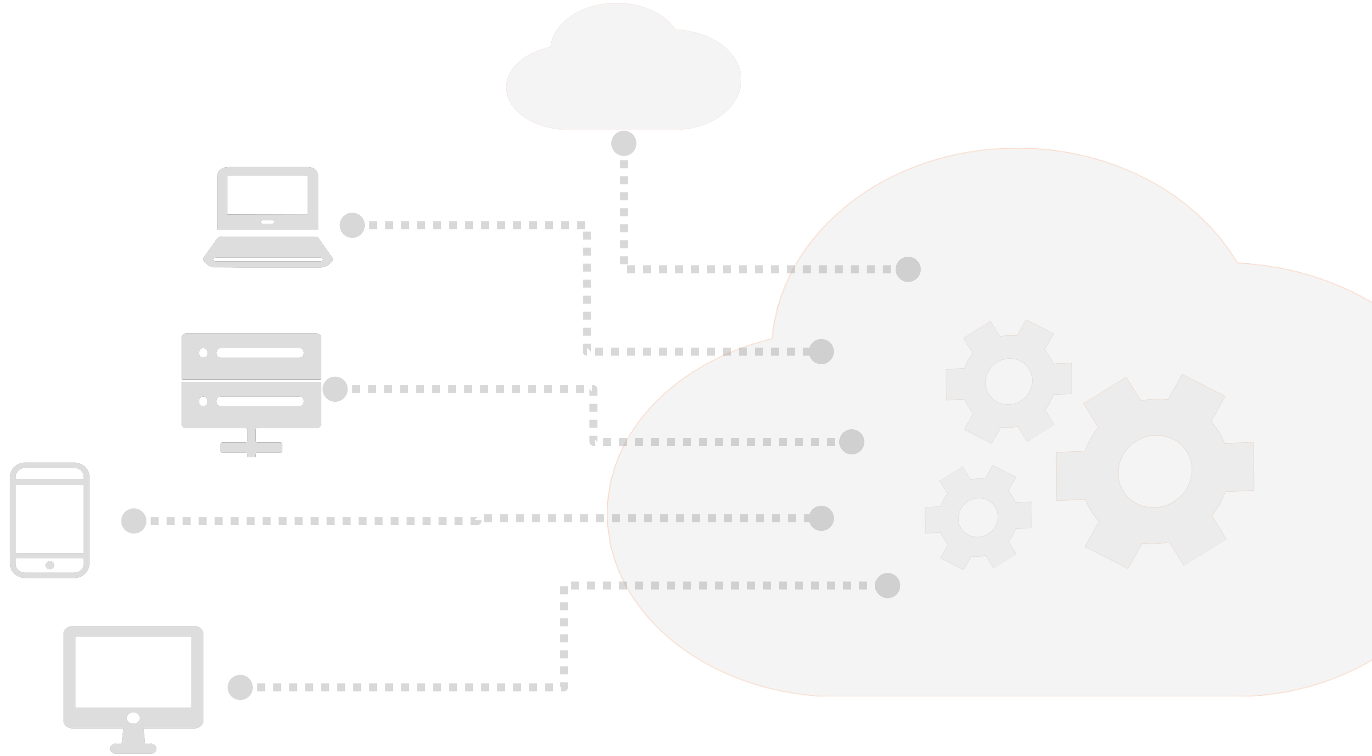


CS 498: Machine Learning Systems

Instructor: Fan Lai



ILLINOIS

About Instructor (Fan)

- **Assistant Professor in CS (2024-present)**
  - PhD’23: "Minimalist Systems for Pervasive Machine Learning’’

- **Research interest**
  - Cloud ML (LLMs, Image/Video Gen)
  - On-device ML, ML4Sys, ML efficiency algs

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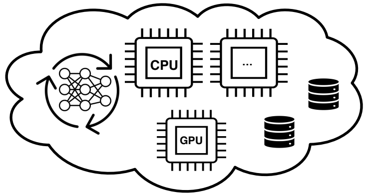

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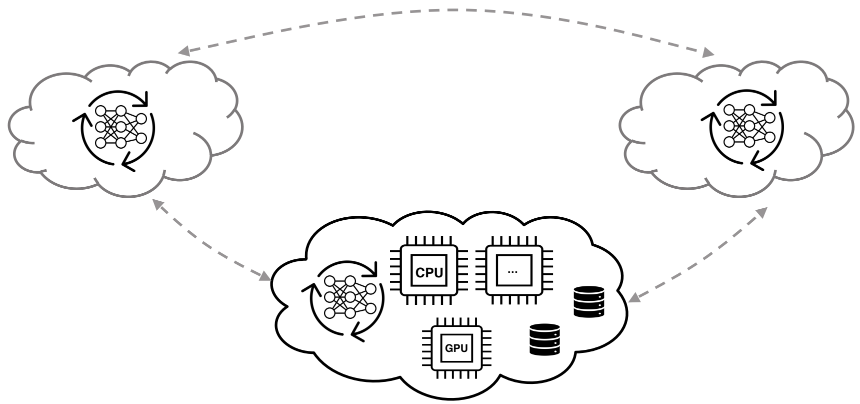

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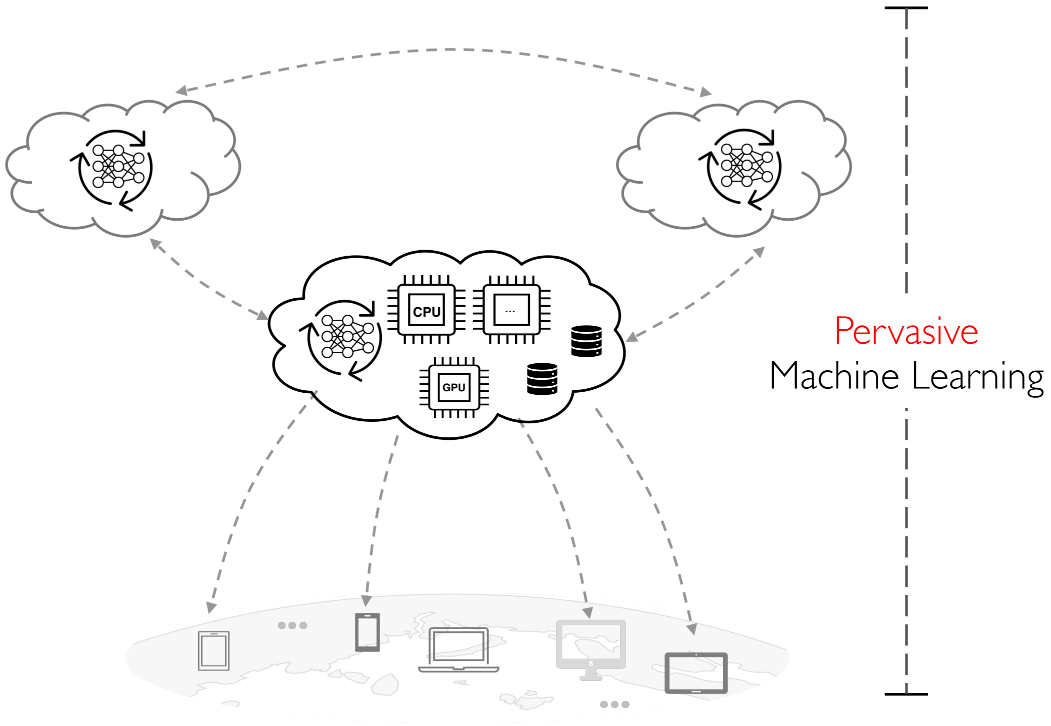

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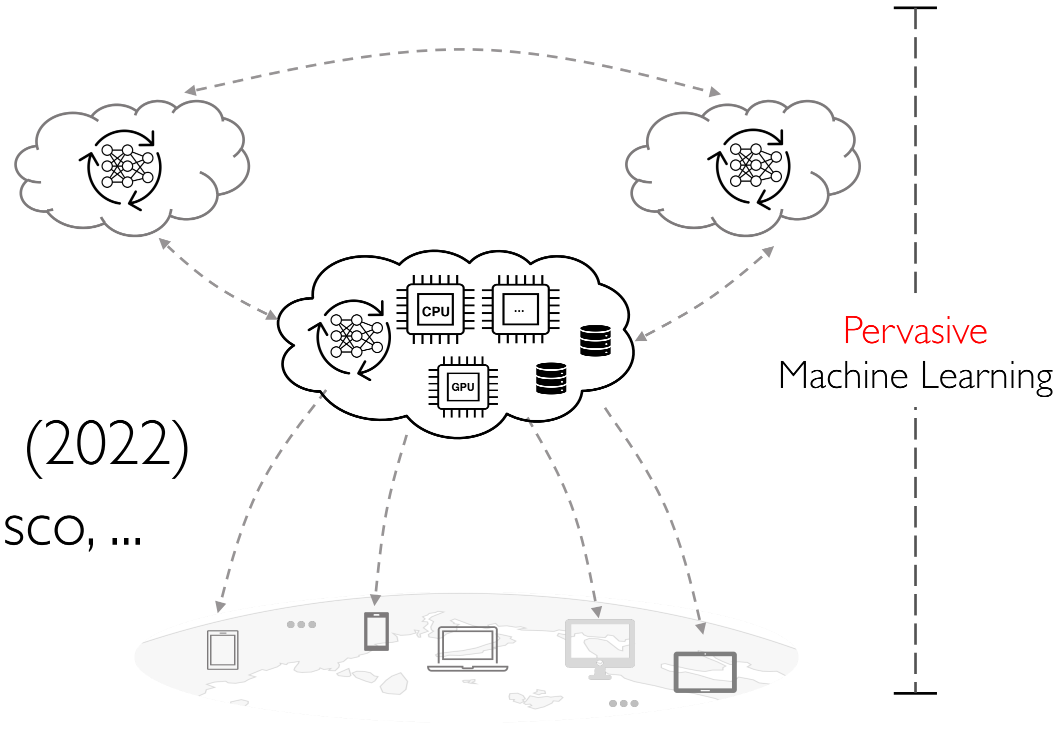
  - PhD'23: "Minimalist Systems for Pervasive Machine Learning"
- **Research interest**
  - Cloud ML (LLMs, Image/Video Gen)
  - On-device ML, ML4Sys, ML efficiency algs
- **Industry**
  - Visiting Faculty@Google (2023-2024), Meta AI (2022)
  - MLSys adoptions at Google, Meta, LinkedIn, Cisco, ...
- **Office hours:** SC 3128, Mon 2 PM – 3 PM



About Our Two TAs (Tony and Banruo)

Tony Hong

(Research: Agent Serving)

Banruo Liu

(Research: Agent Serving)

About Our Two TAs (Tony and Banruo)

Tony Hong

(Research: Agent Serving)

Banruo Liu

(Research: Agent Serving)

Office Hours: Friday 2-3 PM (Zoom, or in-person for assignment due dates)

Status and Prerequisites

- **As of today: ~94 registered**
  - If you are not planning to take the class, please drop ASAP

Status and Prerequisites

- **As of today: ~94 registered**
  - If you are not planning to take the class, please drop ASAP
- **Systems Courses: OS/databases/distributed systems/networking**
  - Equivalent courses are acceptable as well
- **ML/AI course is helpful but not required**
- **Good programming skills** (e.g., PyTorch, TensorFlow, ...)
  - Build systems for course project

Course Schedule

- Webpage: https://github.com/fanlai0990/cs498
- Discussion, Questions: Please Use Canvas
- In-person lectures, discussion
- Feel free to drop us email too :)

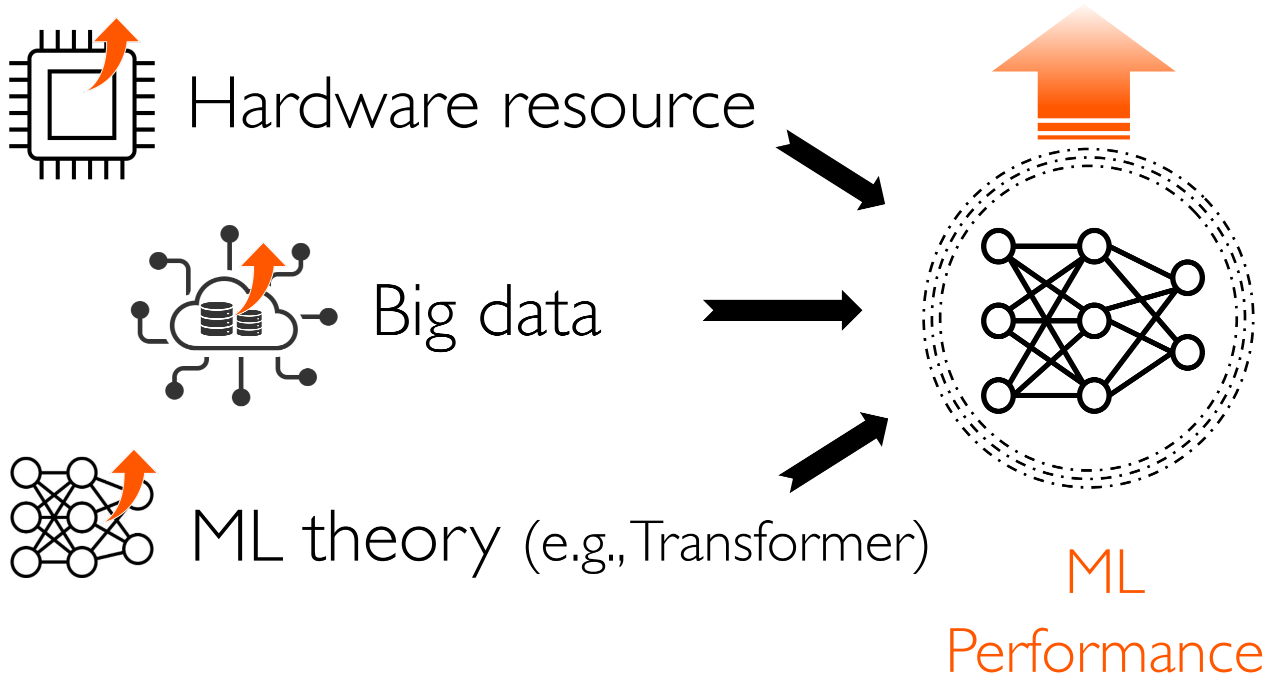
Agenda

- What will you learn in this course?
  - GenAI, systems, networks, ...

- What will you do in this course?
  - Participation, assignment, course project, ...

What Do We Talk About When We Talk About "Machine Learning Systems"

Cloud Enabled Last Leap in Machine Learning (ML)



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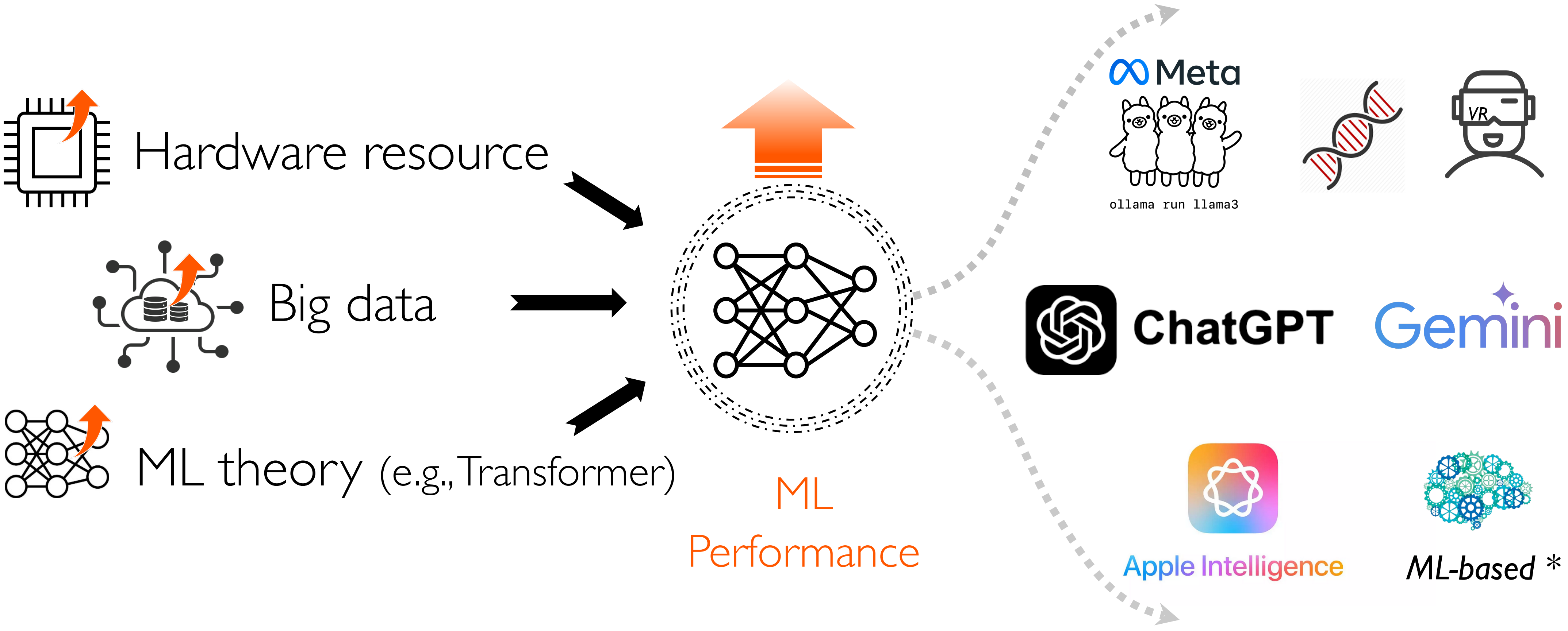
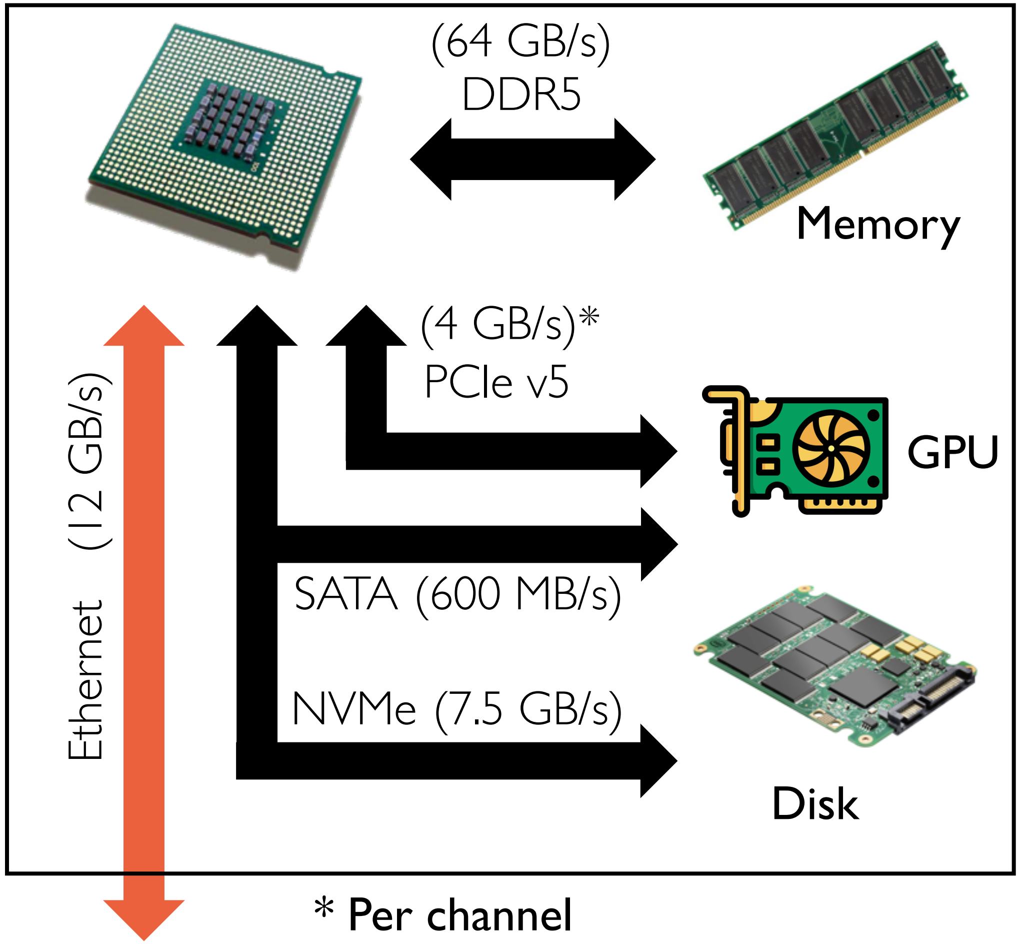


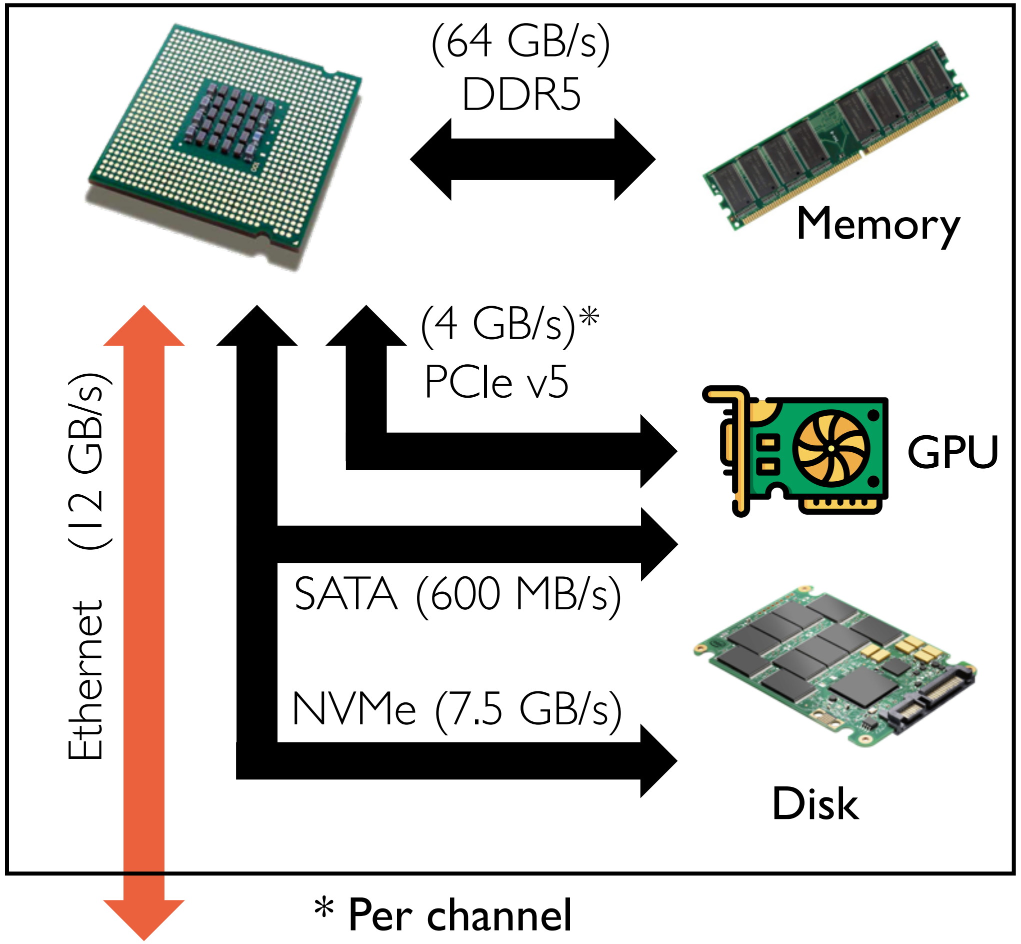
What's in a (Simplified) AI Server?

- **Interconnected compute and storage resources**
  - Different bandwidth and latency constraints

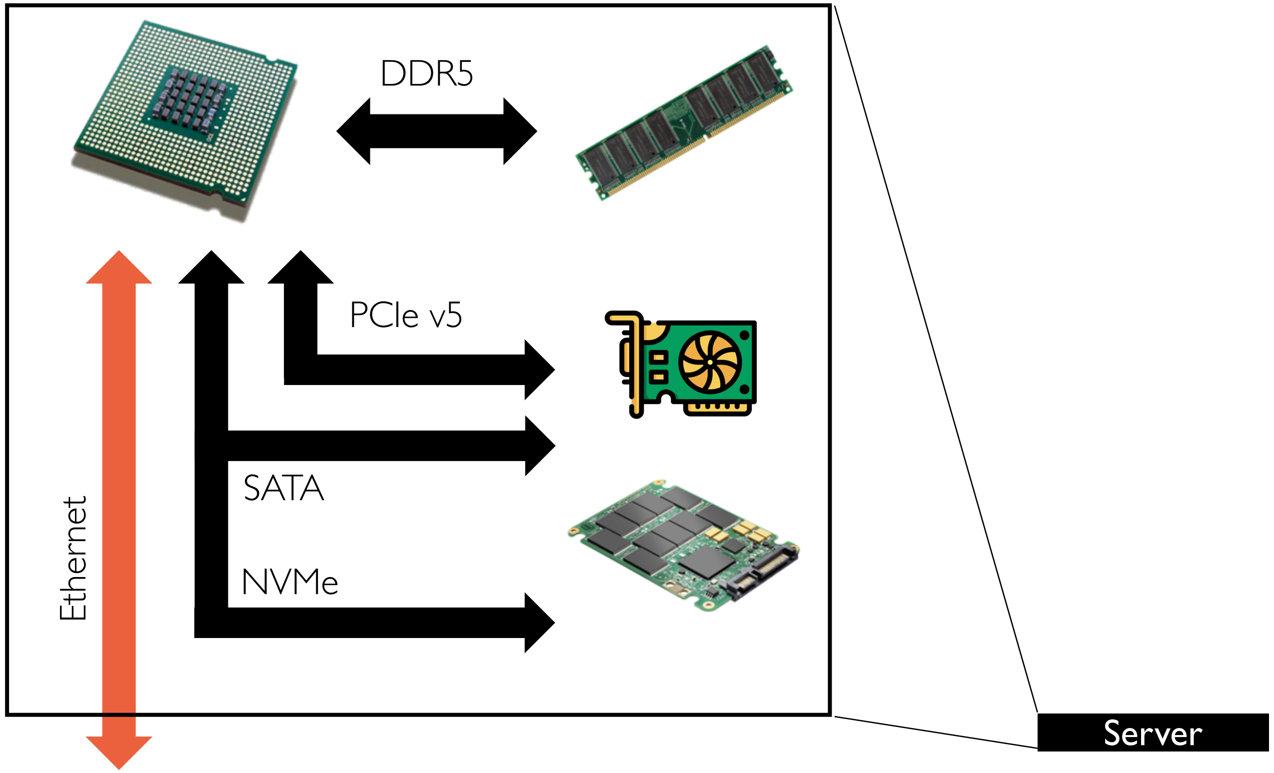


What's in a (Simplified) AI Server?

- **Interconnected compute and storage resources**
  - Different bandwidth and latency constraints
- **Simplified diagram**
  - Doesn't include faster networks such as RDMA, dedicated GPU interconnects such as NVLink



Cloud Datacenter: Scale-Out Servers



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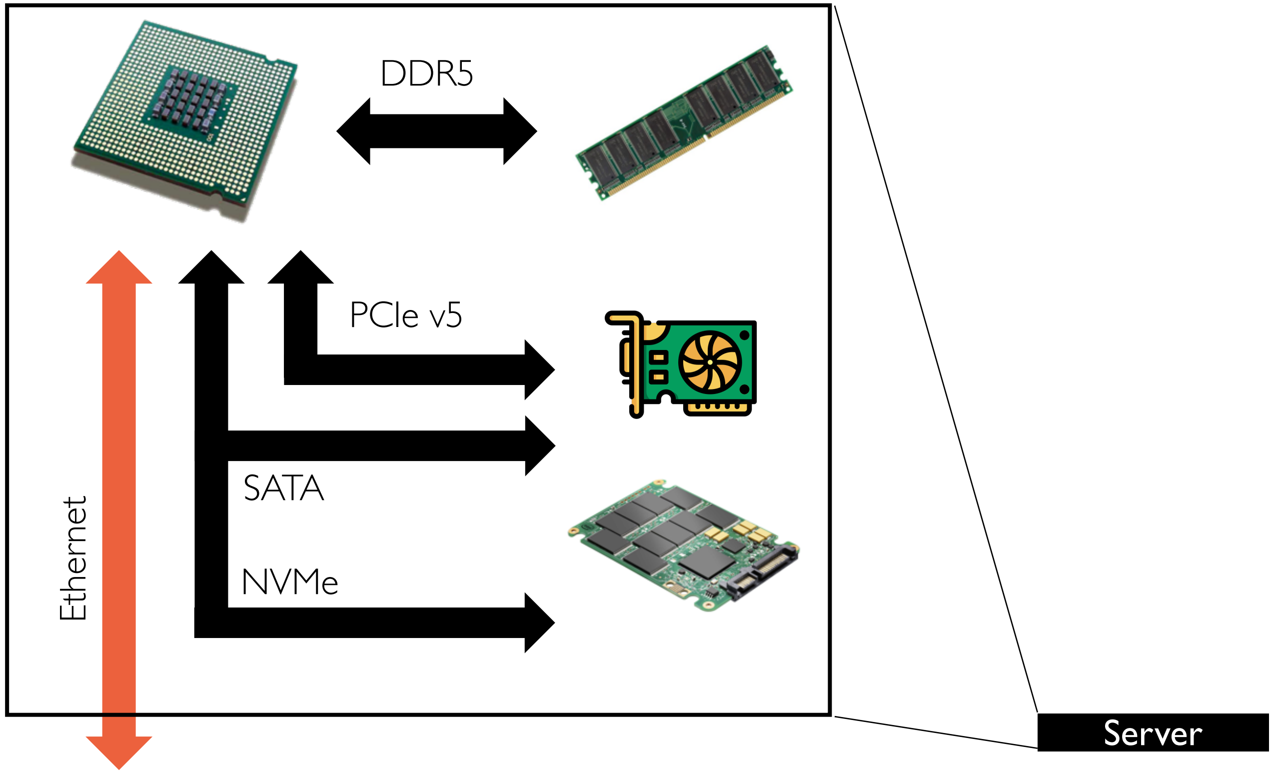
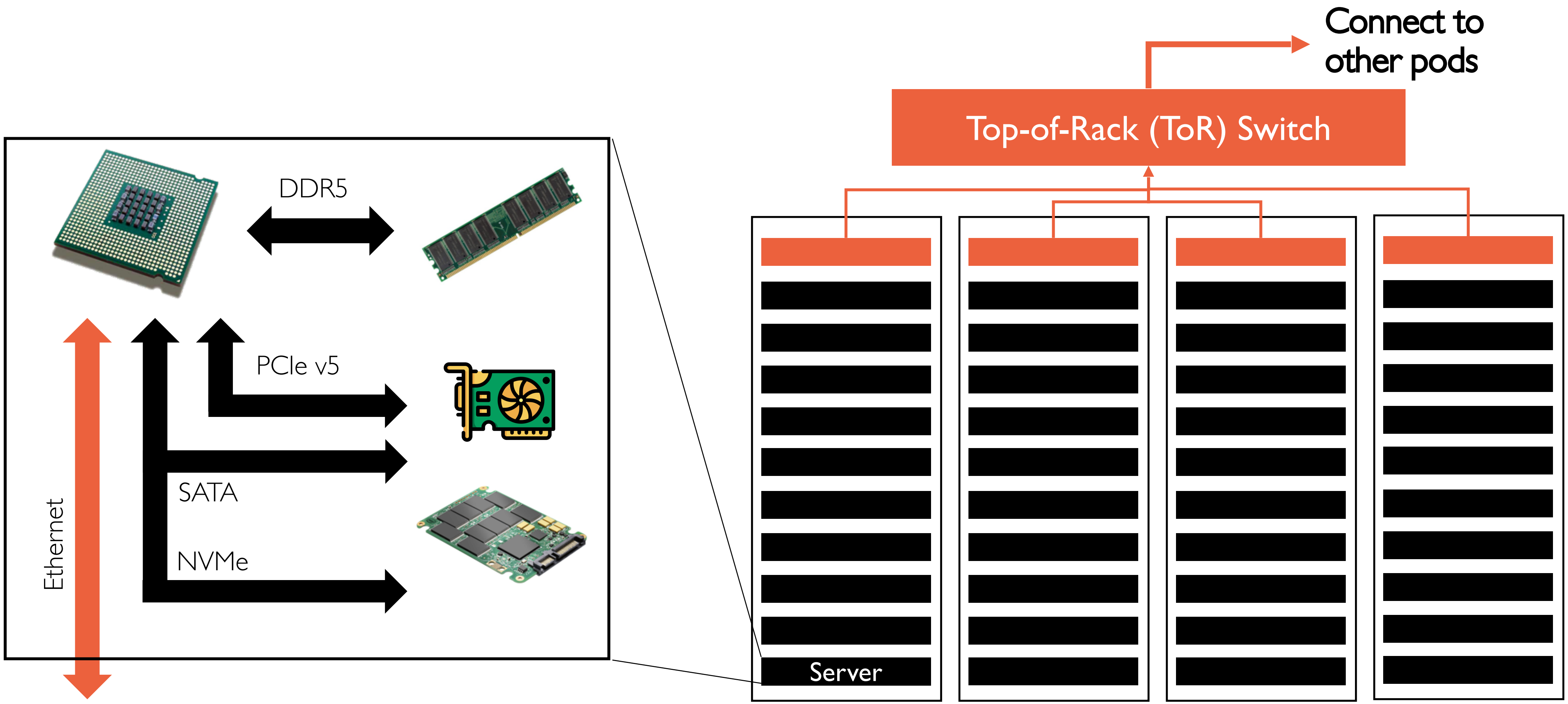


Meta's AI datacenter houses 16K of its 24K GPUs for training the latest Llama 3 model

Cloud Datacenter: Scale-Out Servers



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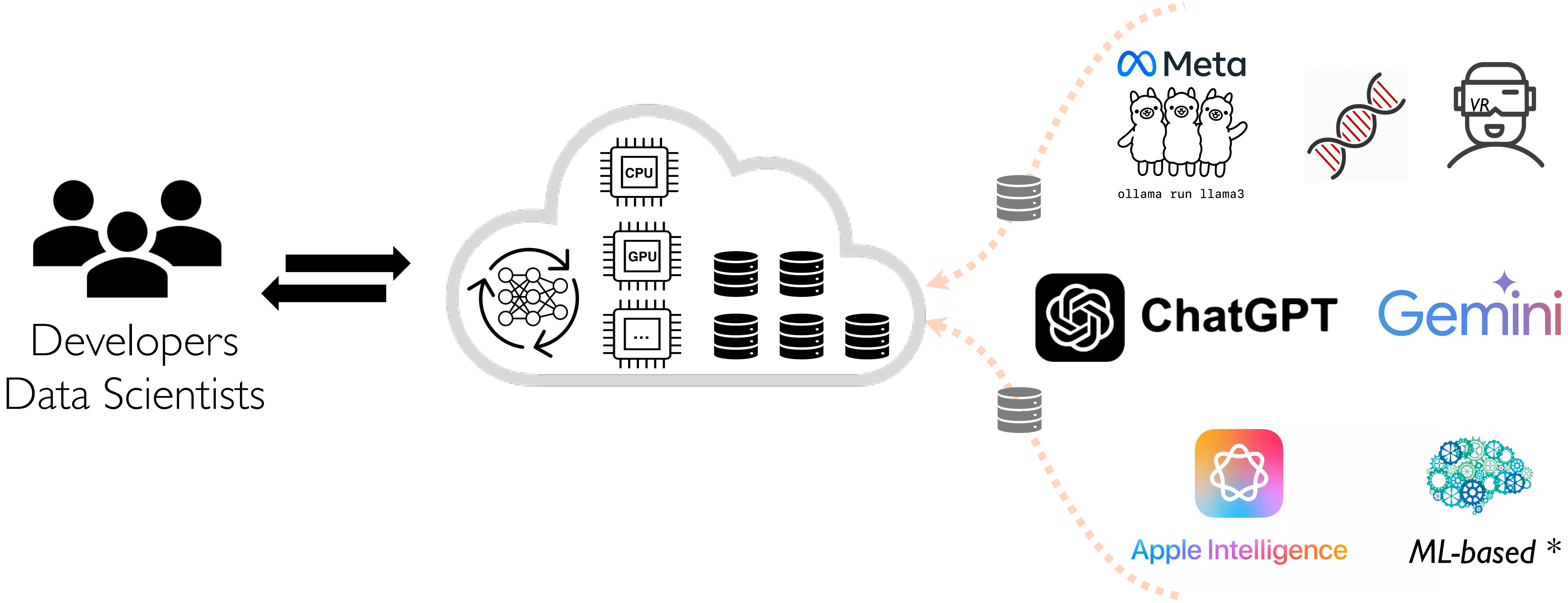
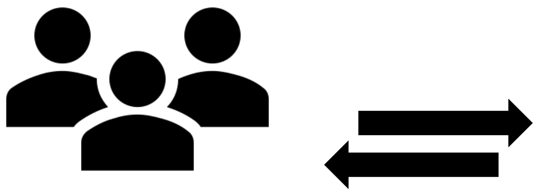
Cloud Faces Skyrocketing ML Workloads

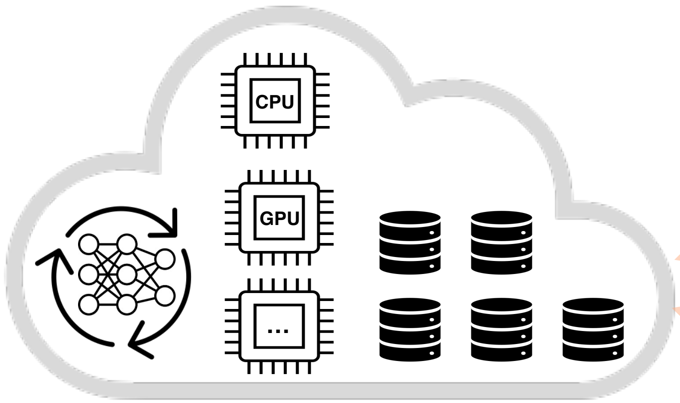


Cloud Faces Skyrocketing ML Workloads

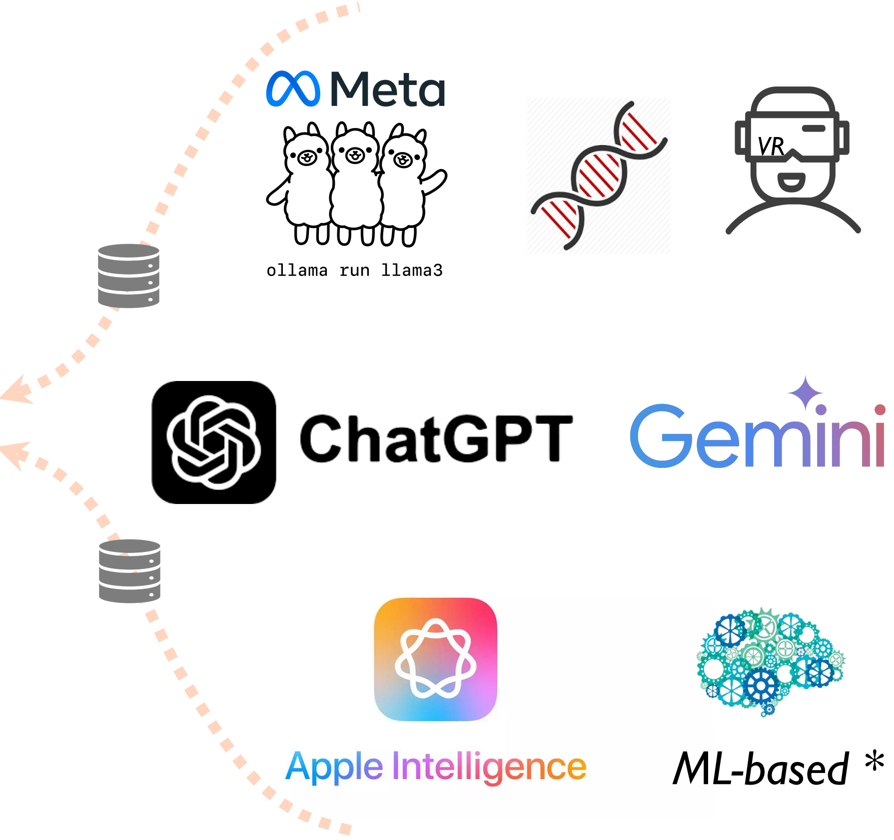
Model size **doubled**
every two years [1]



Developers
Data Scientists



Petabytes of data
ingestion every day [1]



[1] "Sustainable AI: Environment Implications, Challenges and Opportunities", Meta, MLSys'22

GenAI Lifecycle Demands Great Systems Support

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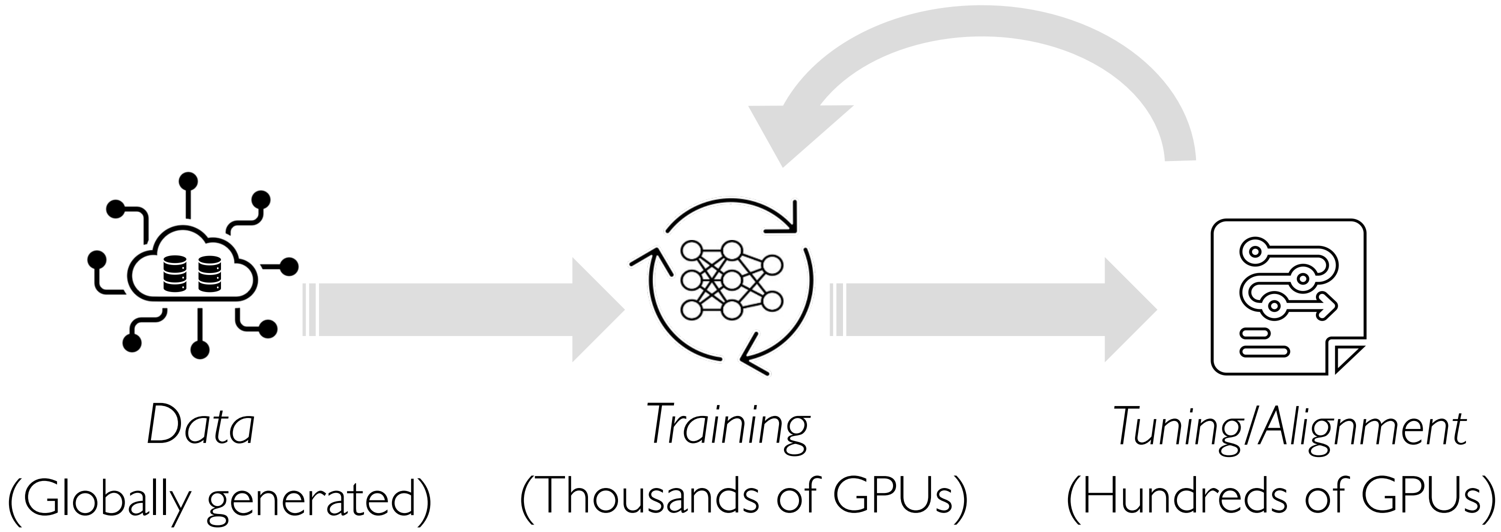

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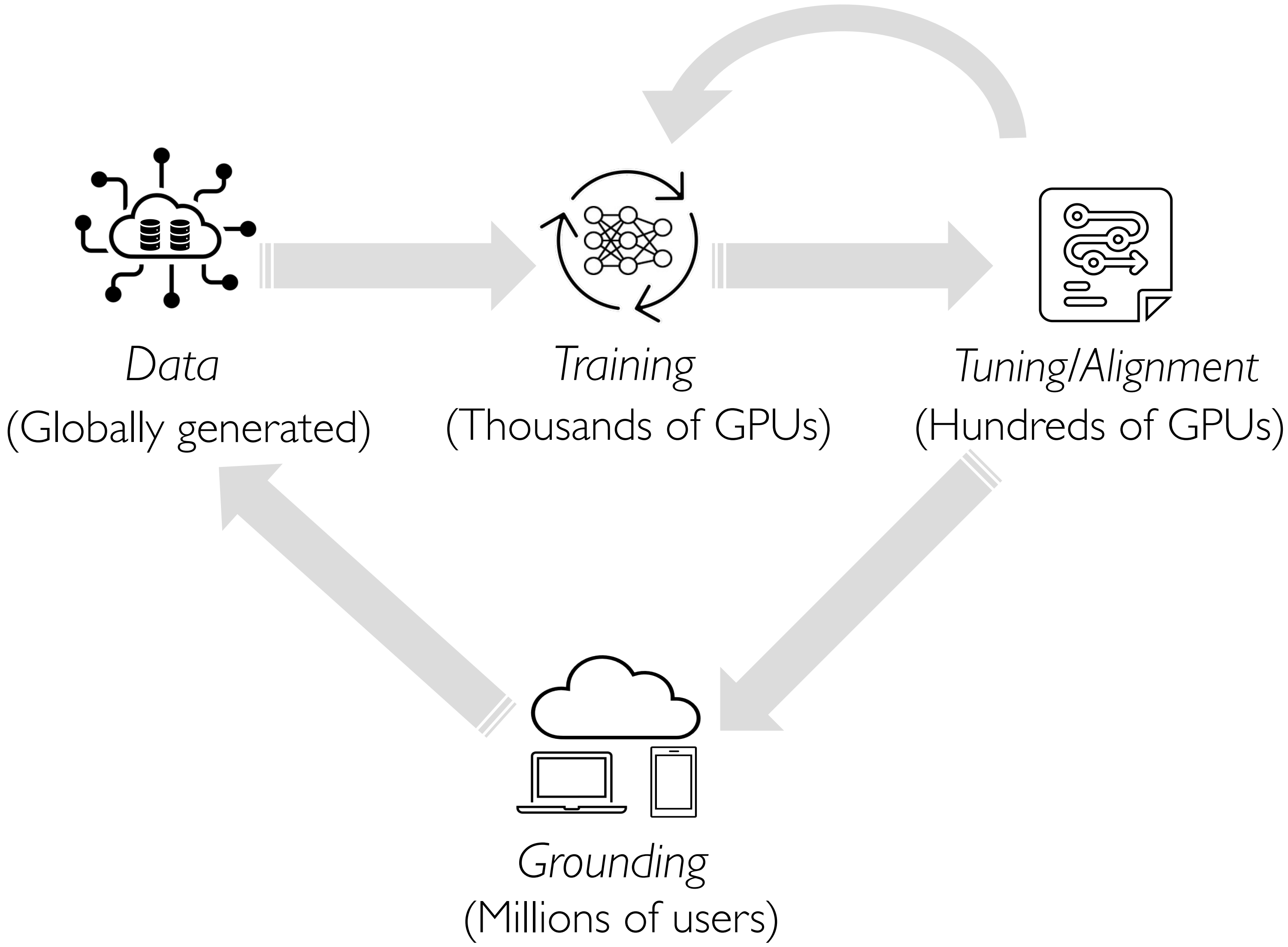

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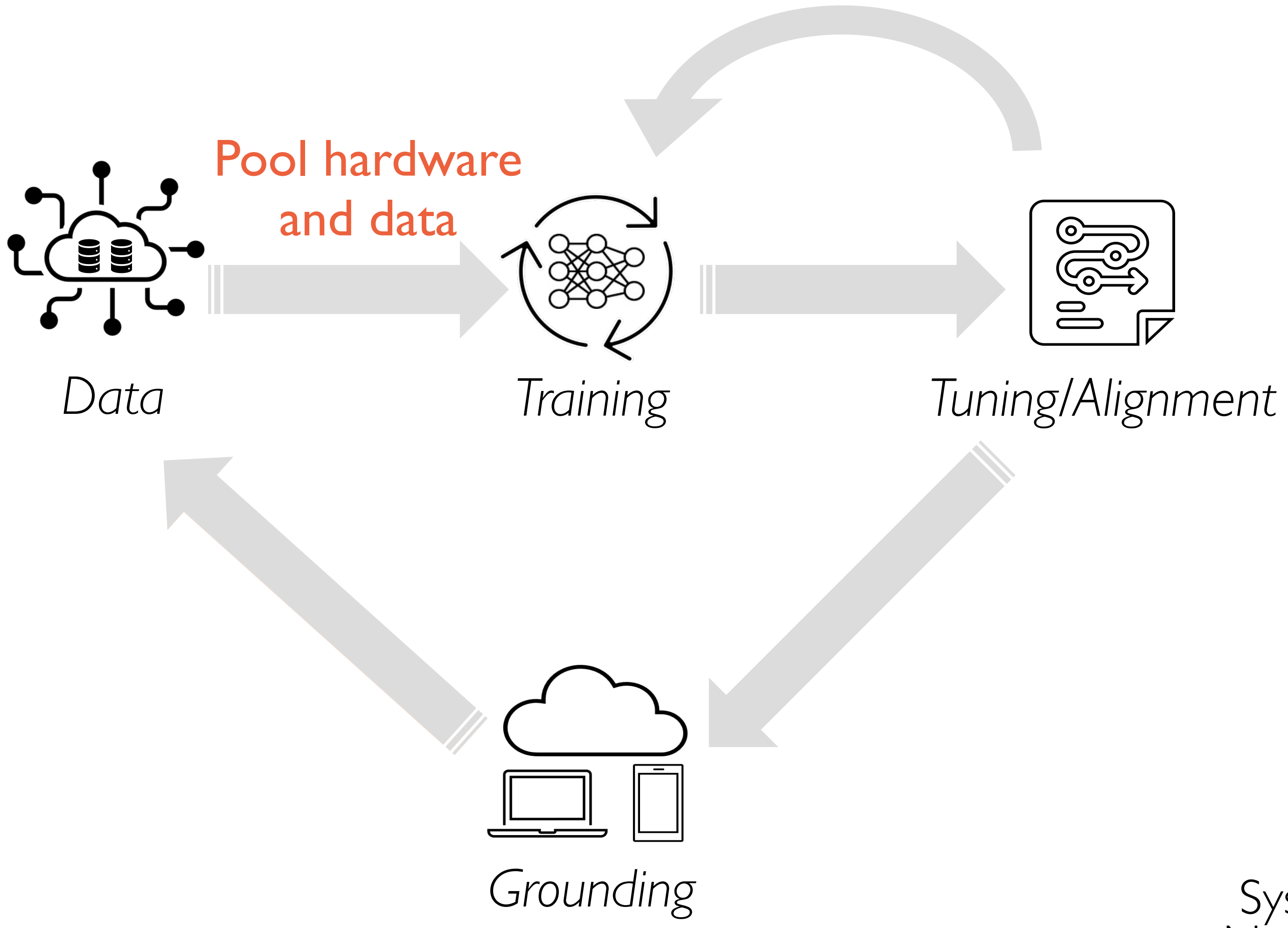
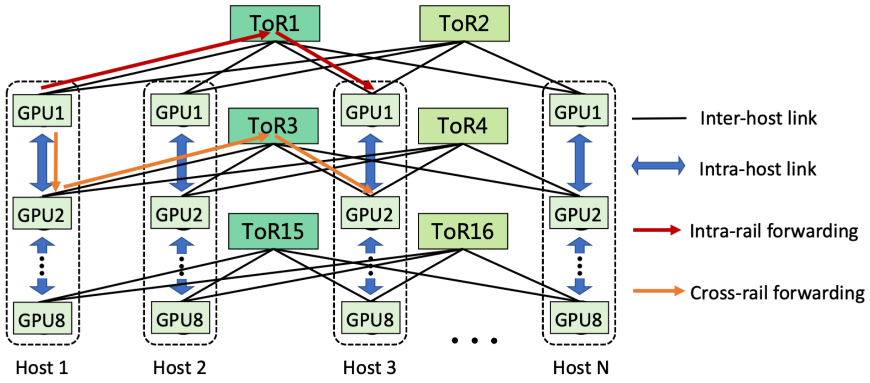


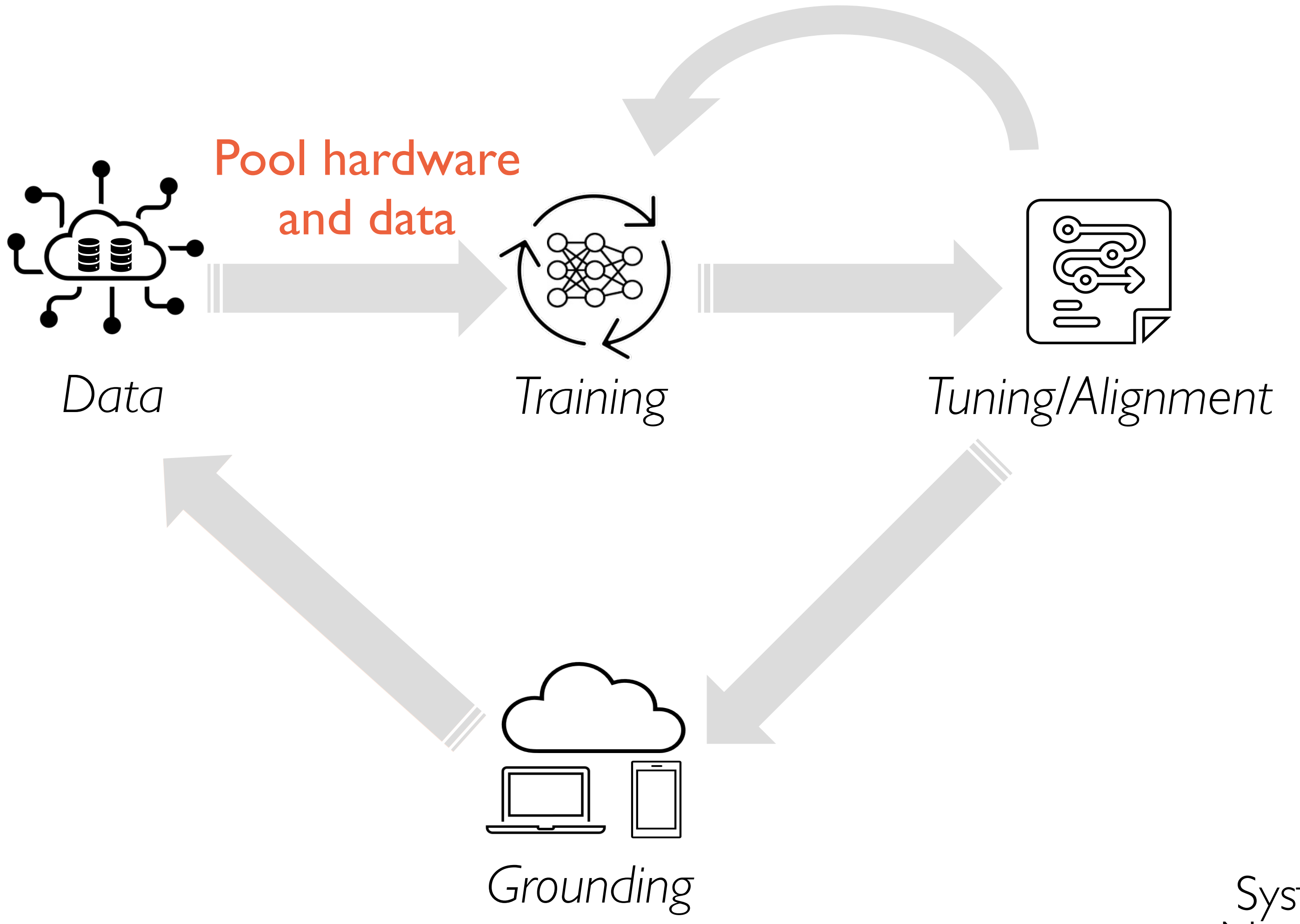
ML



HyperScale Training Infra
(e.g., compute, communication, memory)

Systems &
Networking

GenAI Lifecycle Demands Great Systems Support



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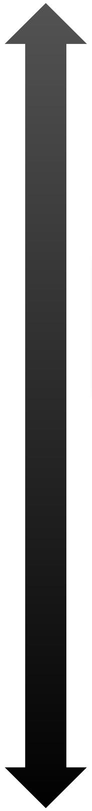

Systems &
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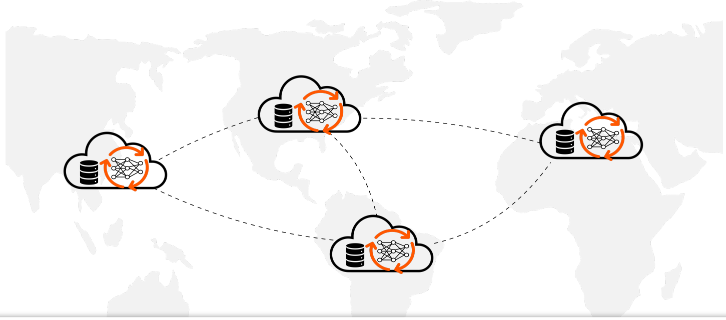

Distributed model training across massive servers and even datacenters

Model Auto-Sharding

HyperScale Training Infra
(e.g., compute, communication, memory)

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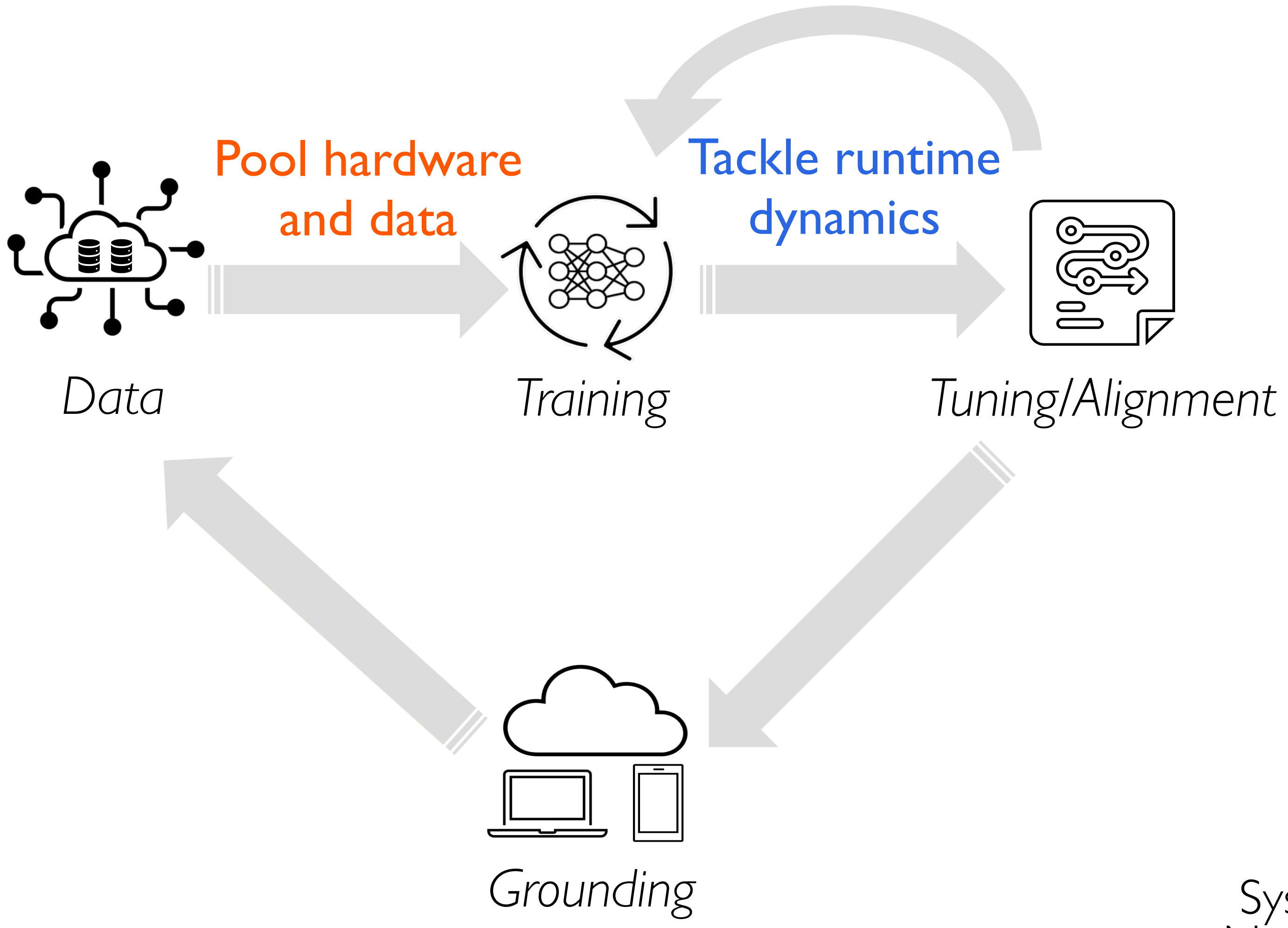

ML

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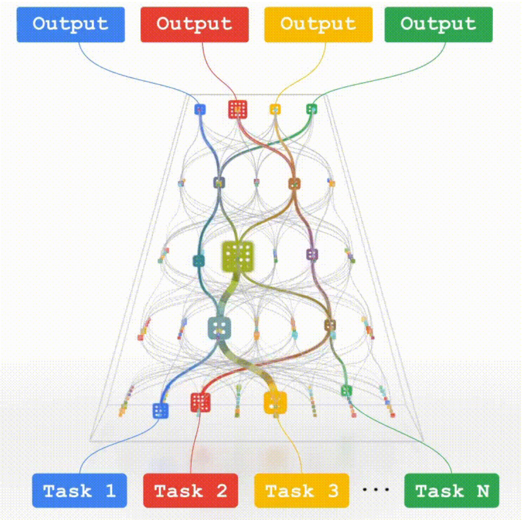

Model Auto-Sharding

Adaptive Runtime

Training can dynamically activate model components (e.g., MoE, sparsity)

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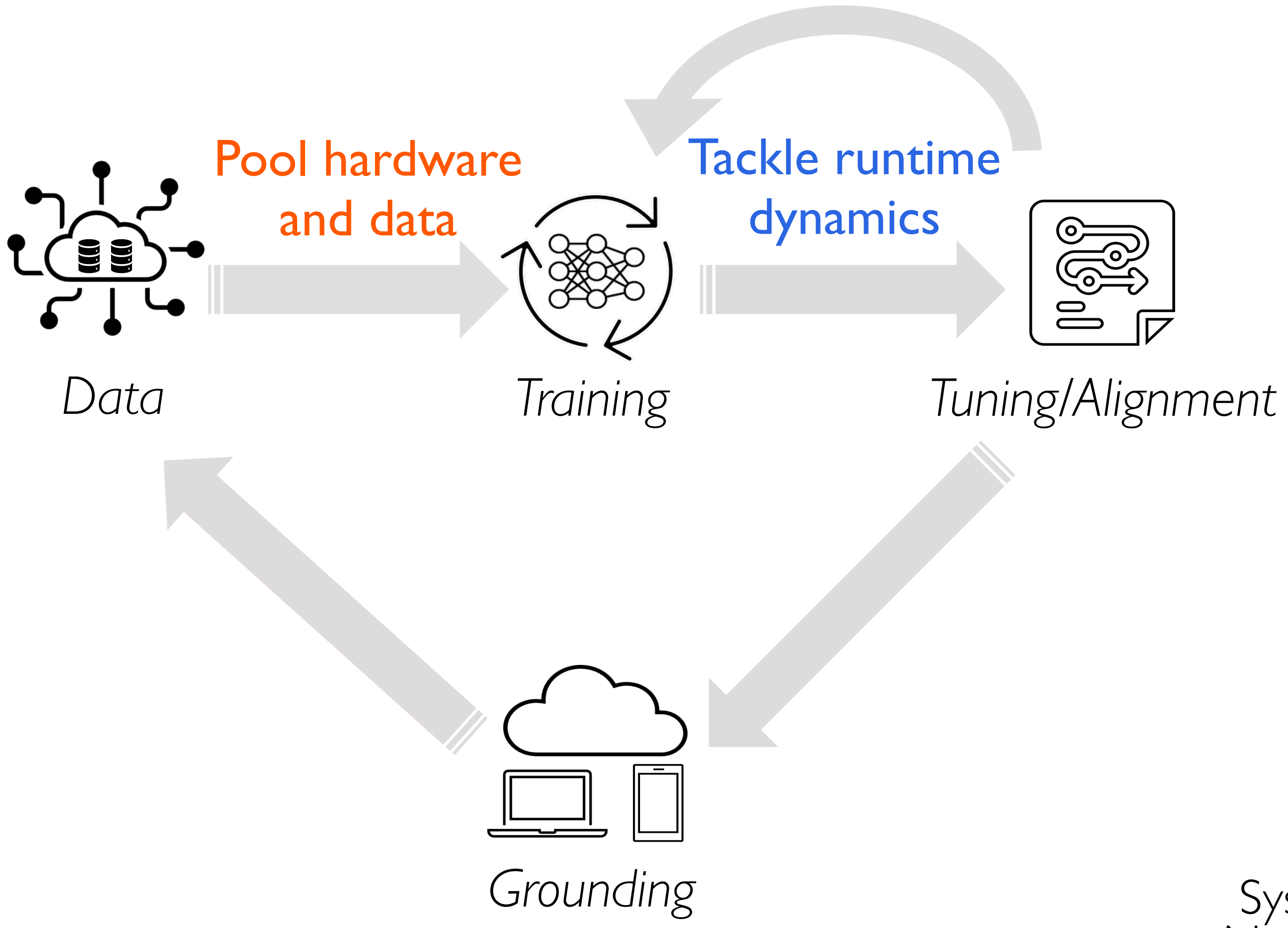

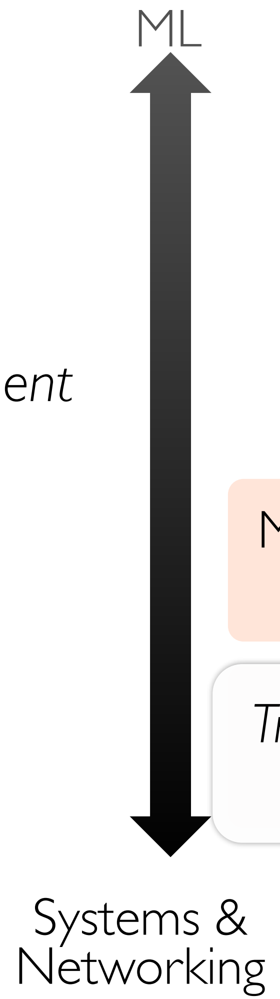

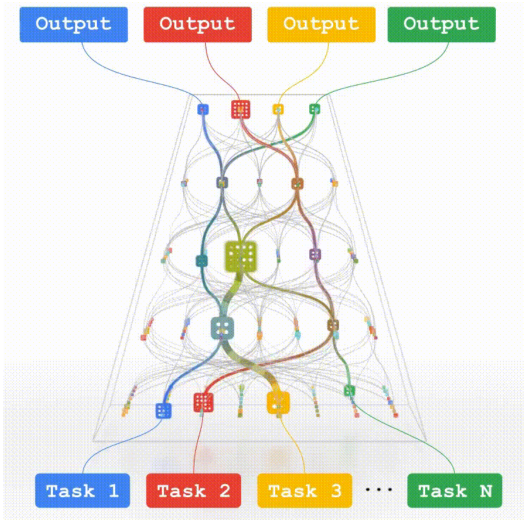

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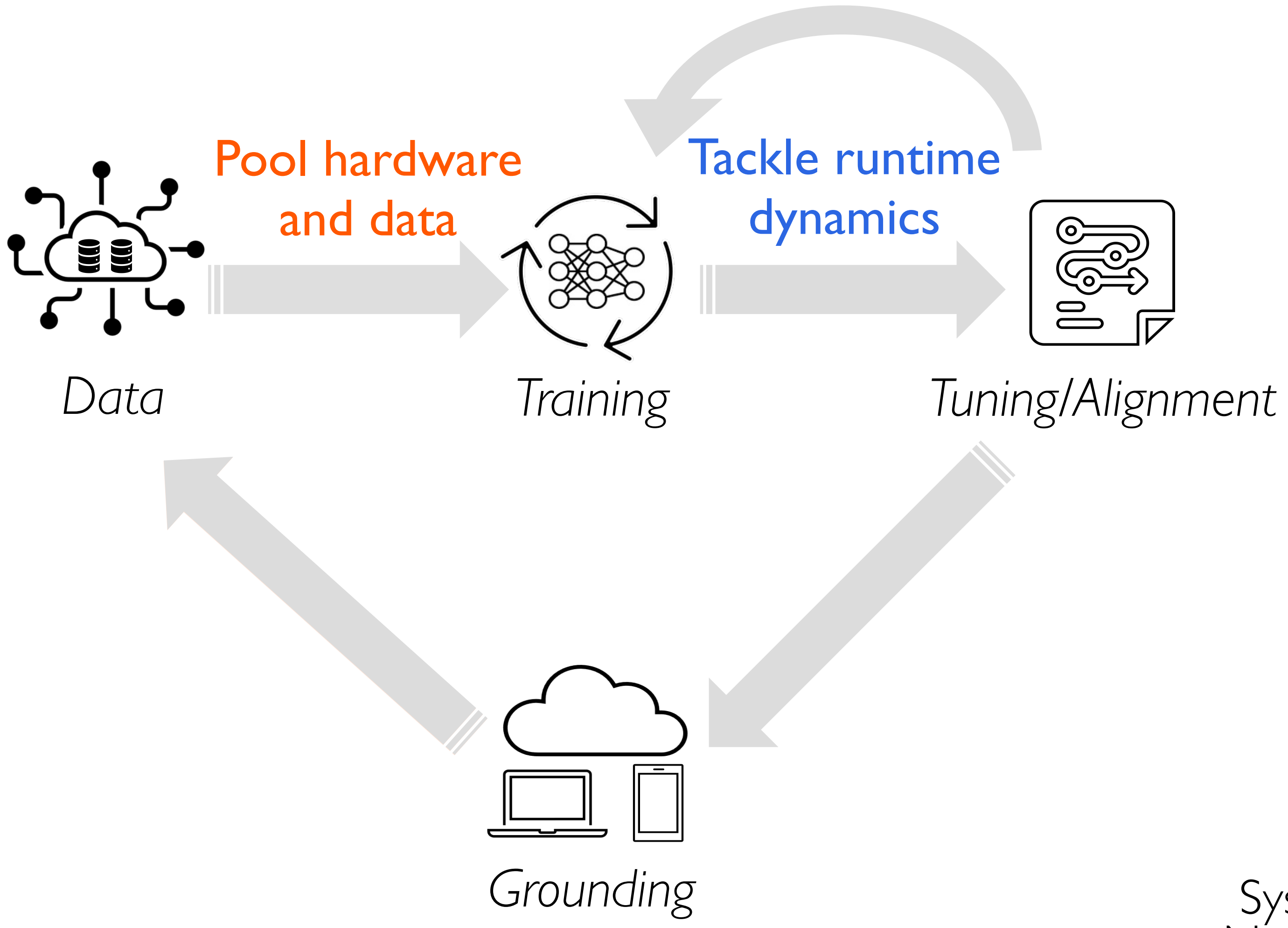

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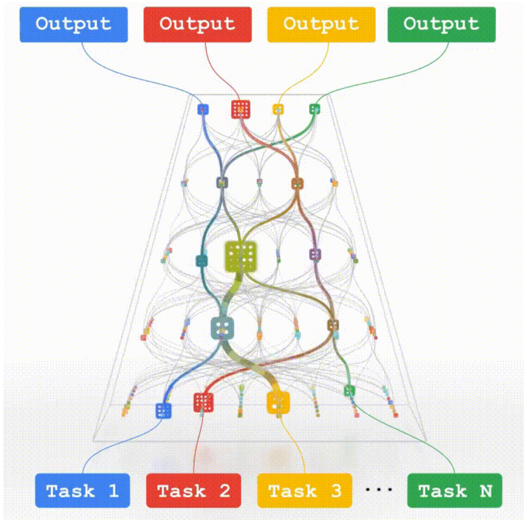


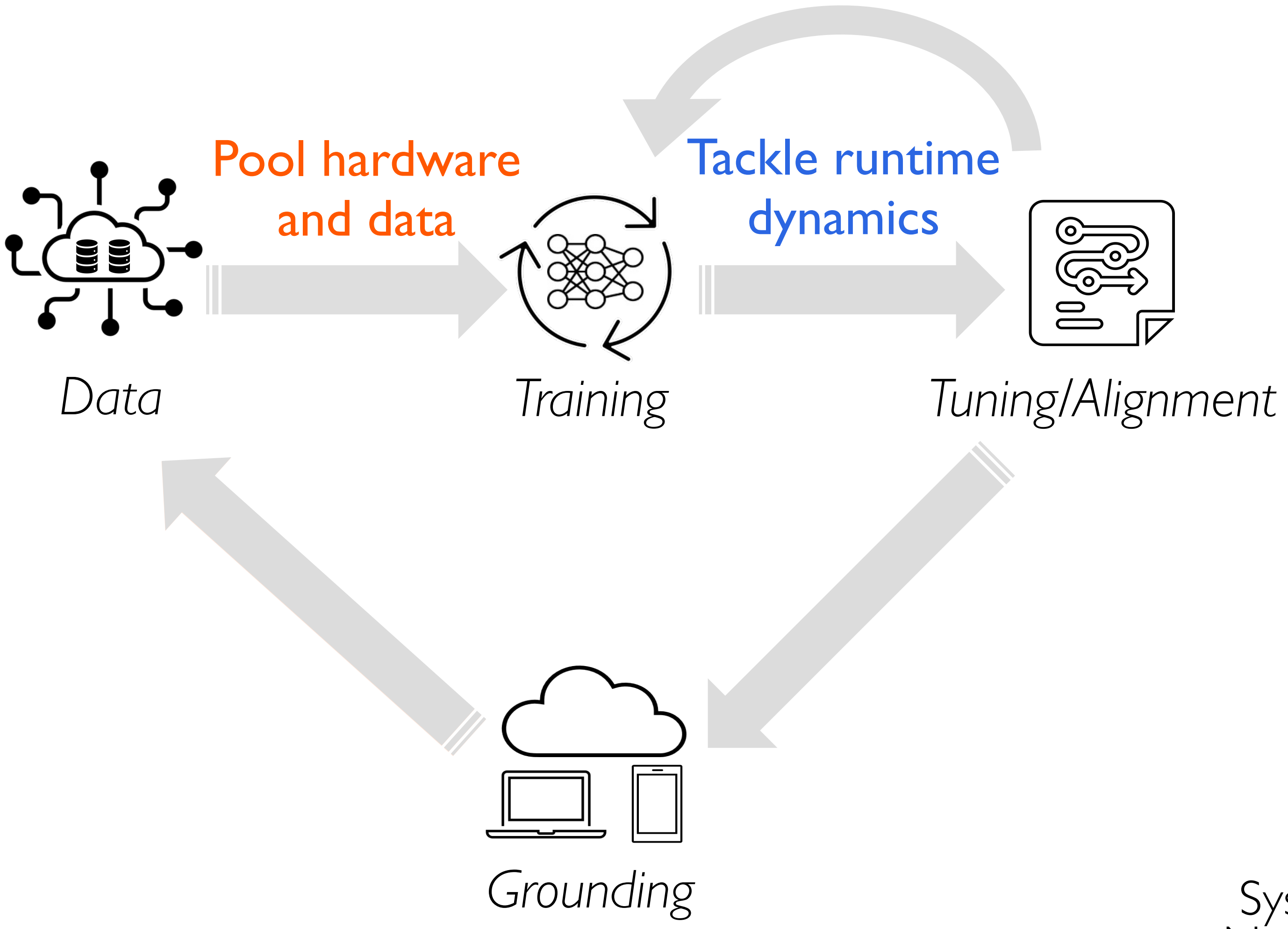
Model Auto-Sharding

Adaptive Runtime

Multi-job Scheduling

Jobs can compete for the shared resources

GenAI Lifecycle Demands Great Systems Support



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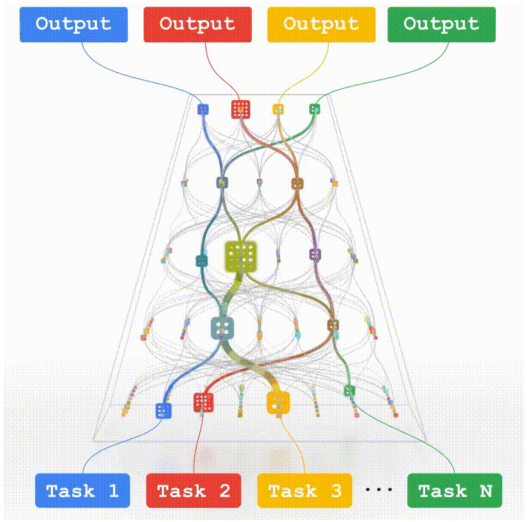

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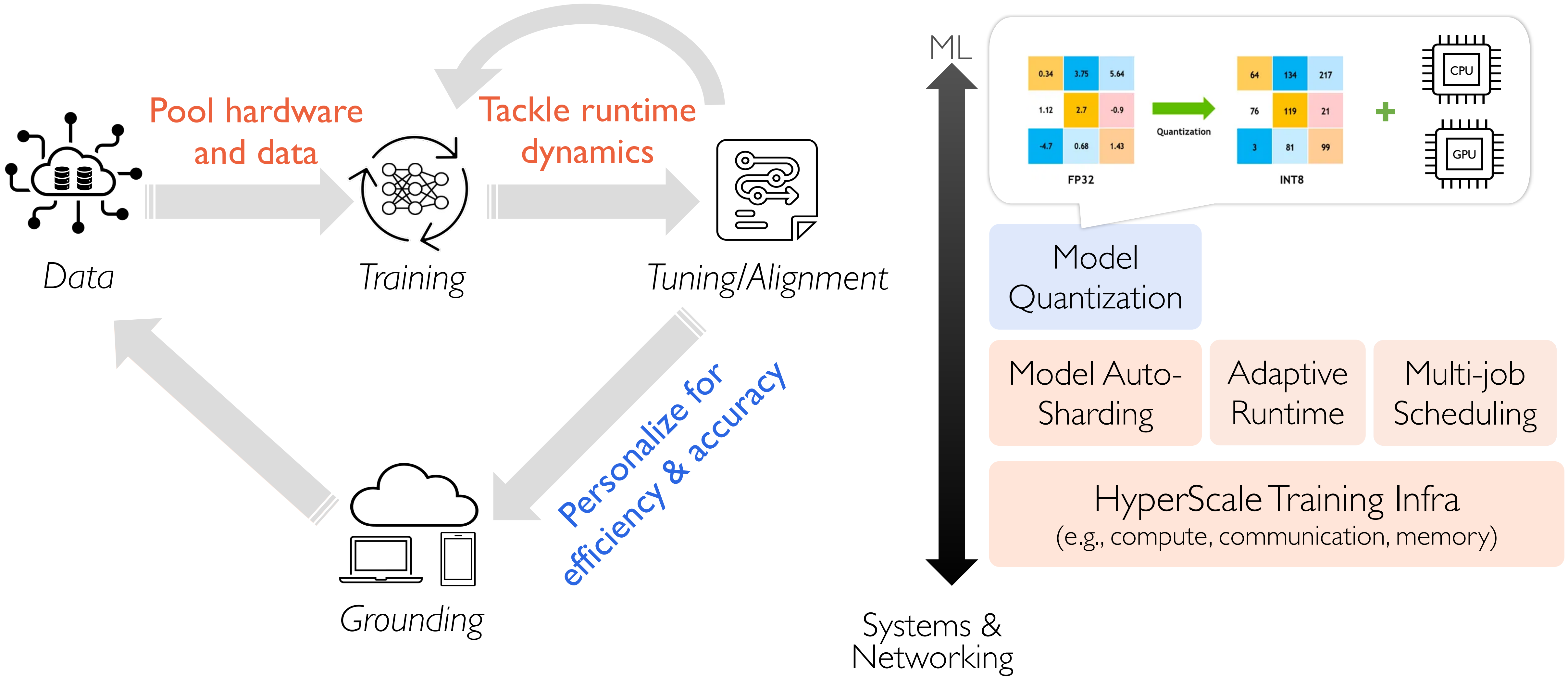

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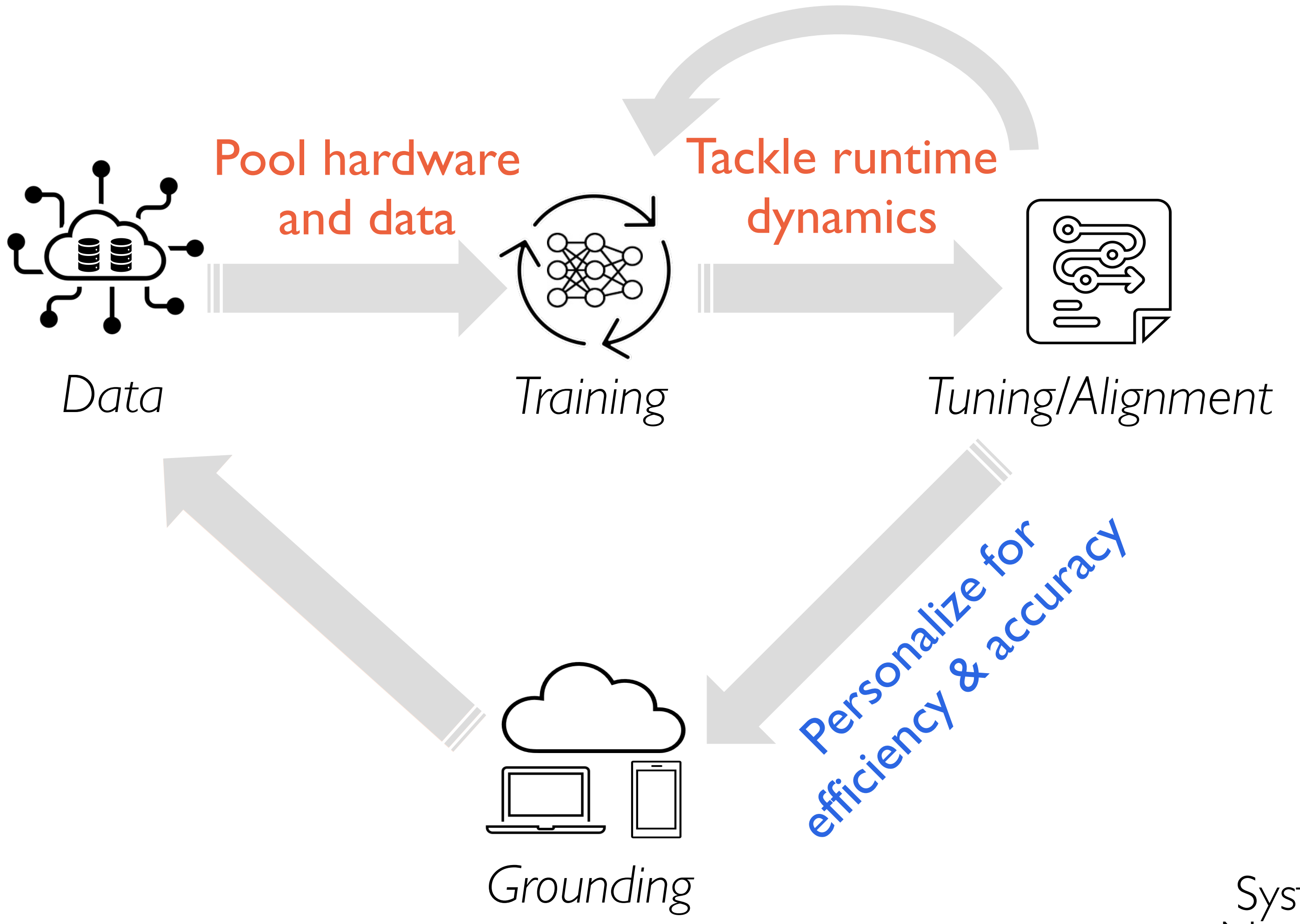

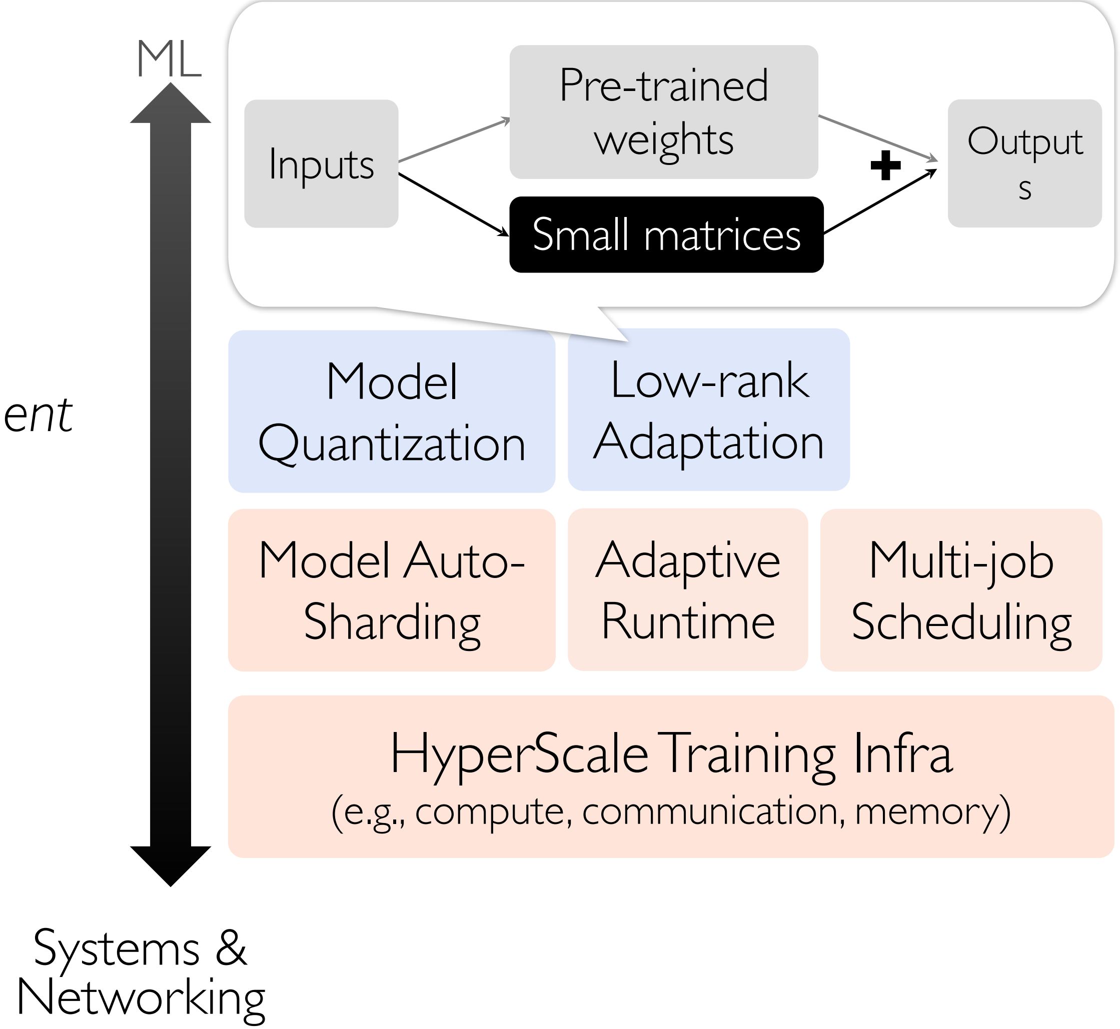

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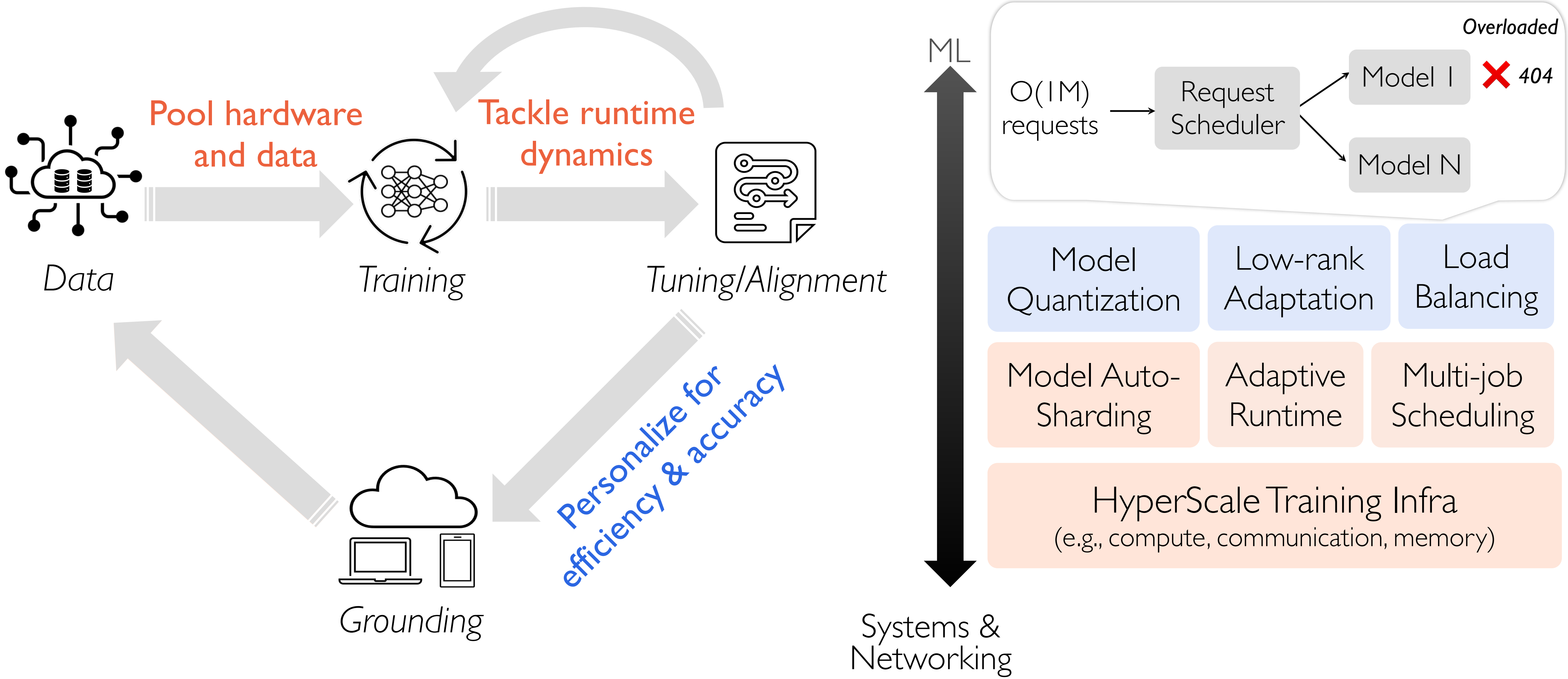

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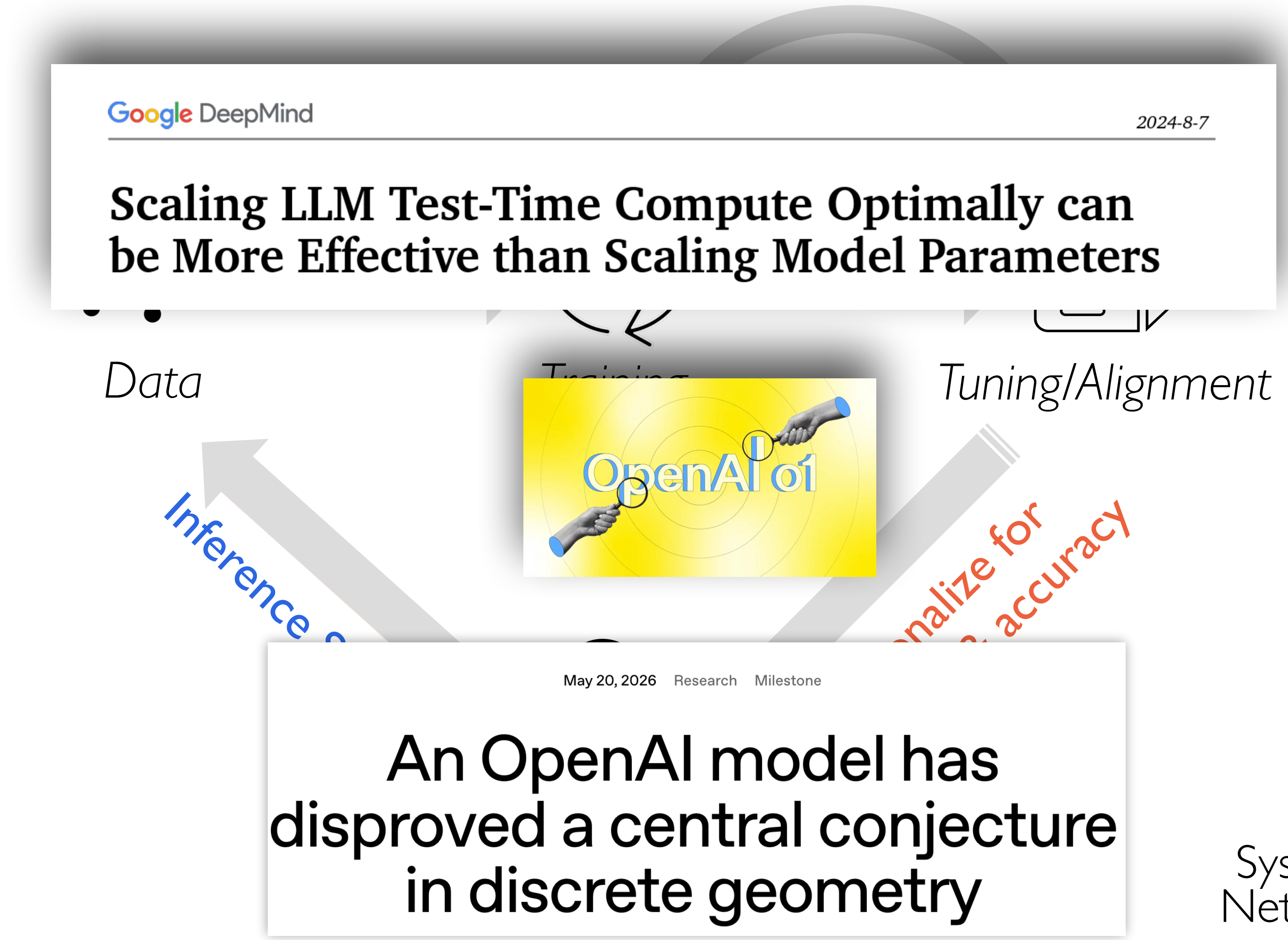


ML

GenAI for X (e.g., systems)

Scaling Reasoning

Model Quantization

Low-rank Adaptation

Load Balancing

Model Auto-Sharding

Adaptive Runtime

Multi-job Scheduling

HyperScale Training Infra
(e.g., compute, communication, memory)

Systems & Networking

What will you learn in this course?

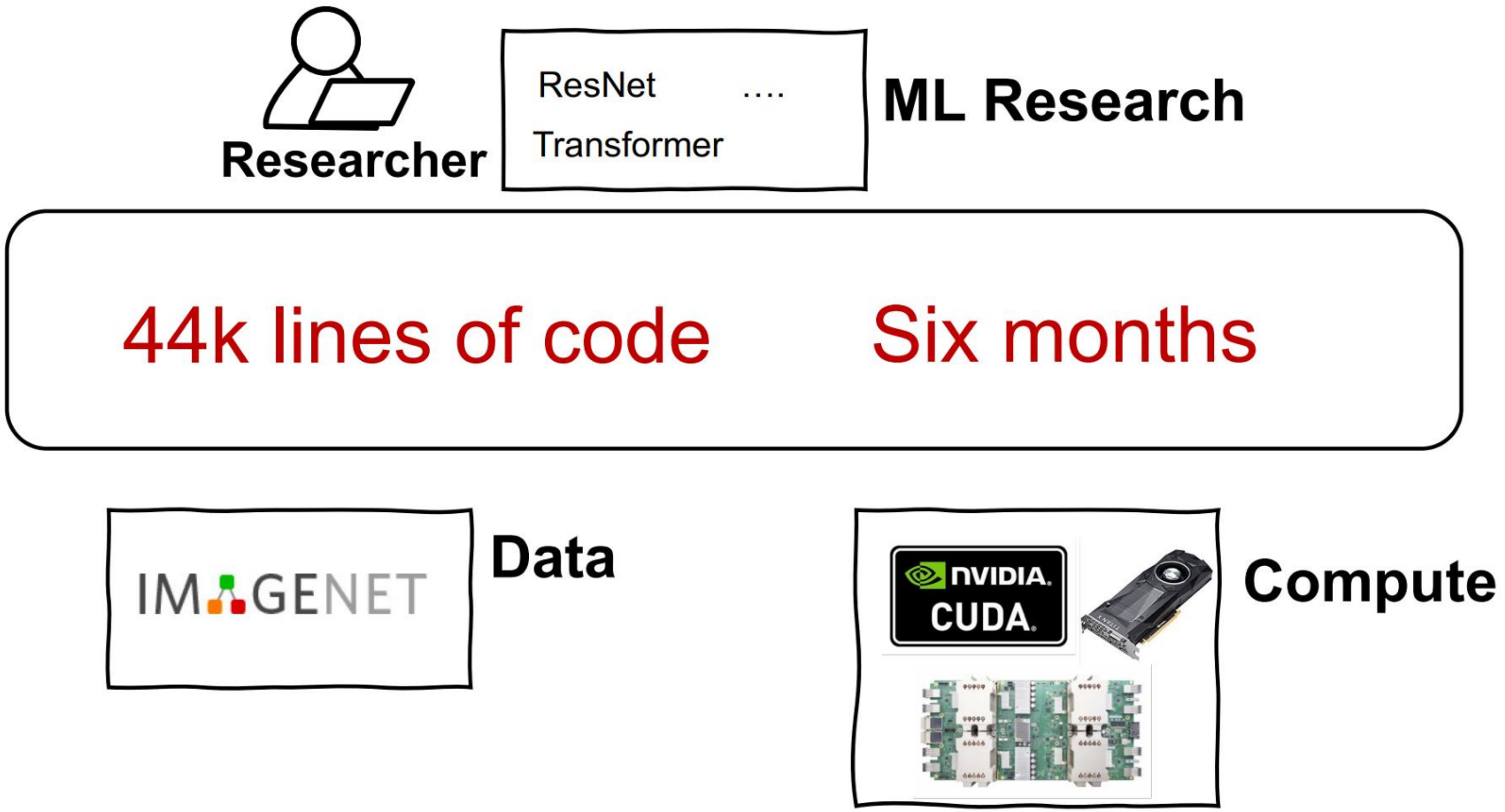
-- How to Support Practical ML

Efficiency: Efficient use of hardware for high scalability and throughput

Effectiveness: High accuracy and fast convergence

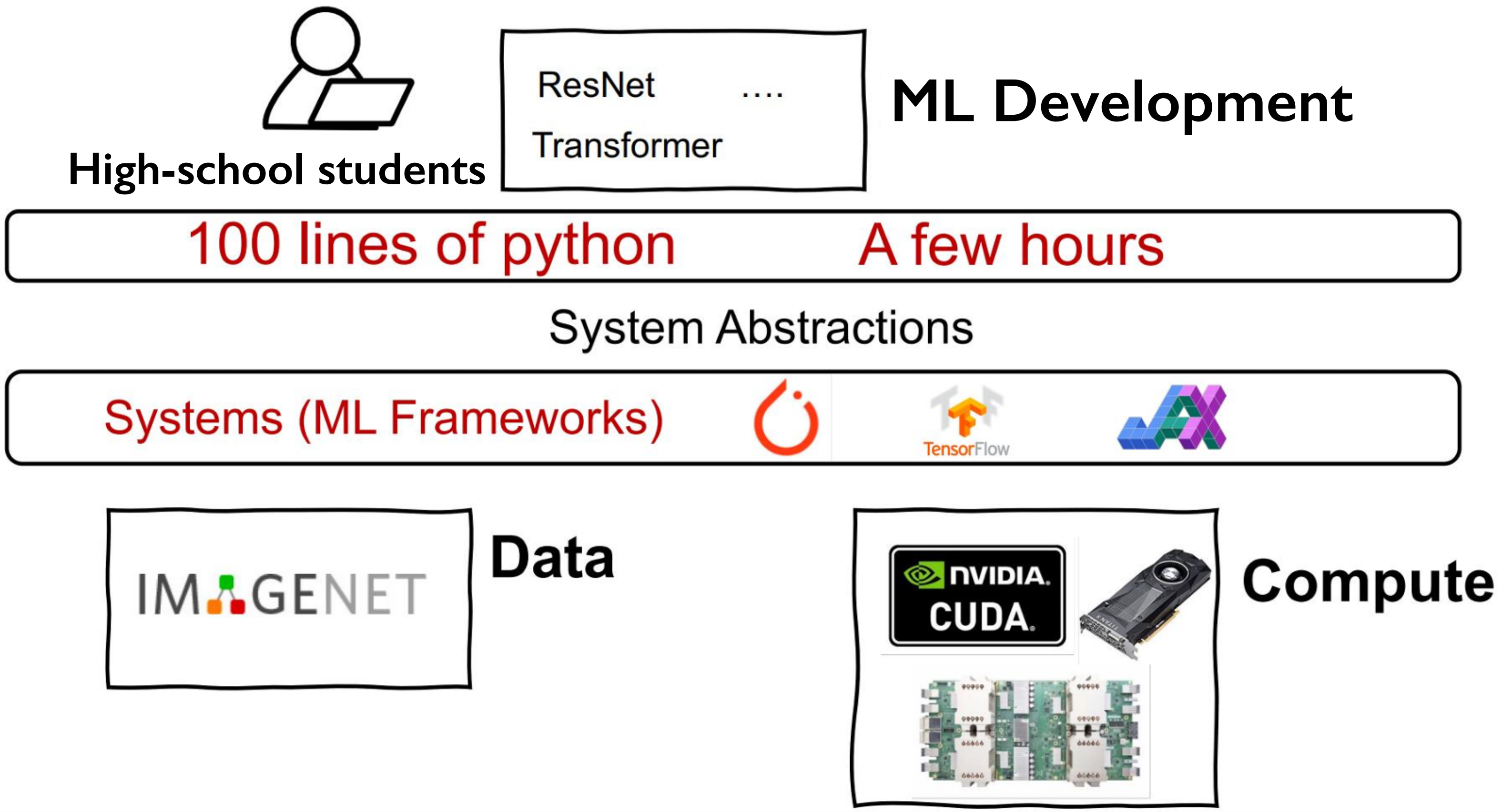
Easy to use: Improve development productivity for developers and users

Without ML Systems



Source: https://mlsyscourse.org/slides/01-course-introduction.pdf

As ML Systems Democratize AI



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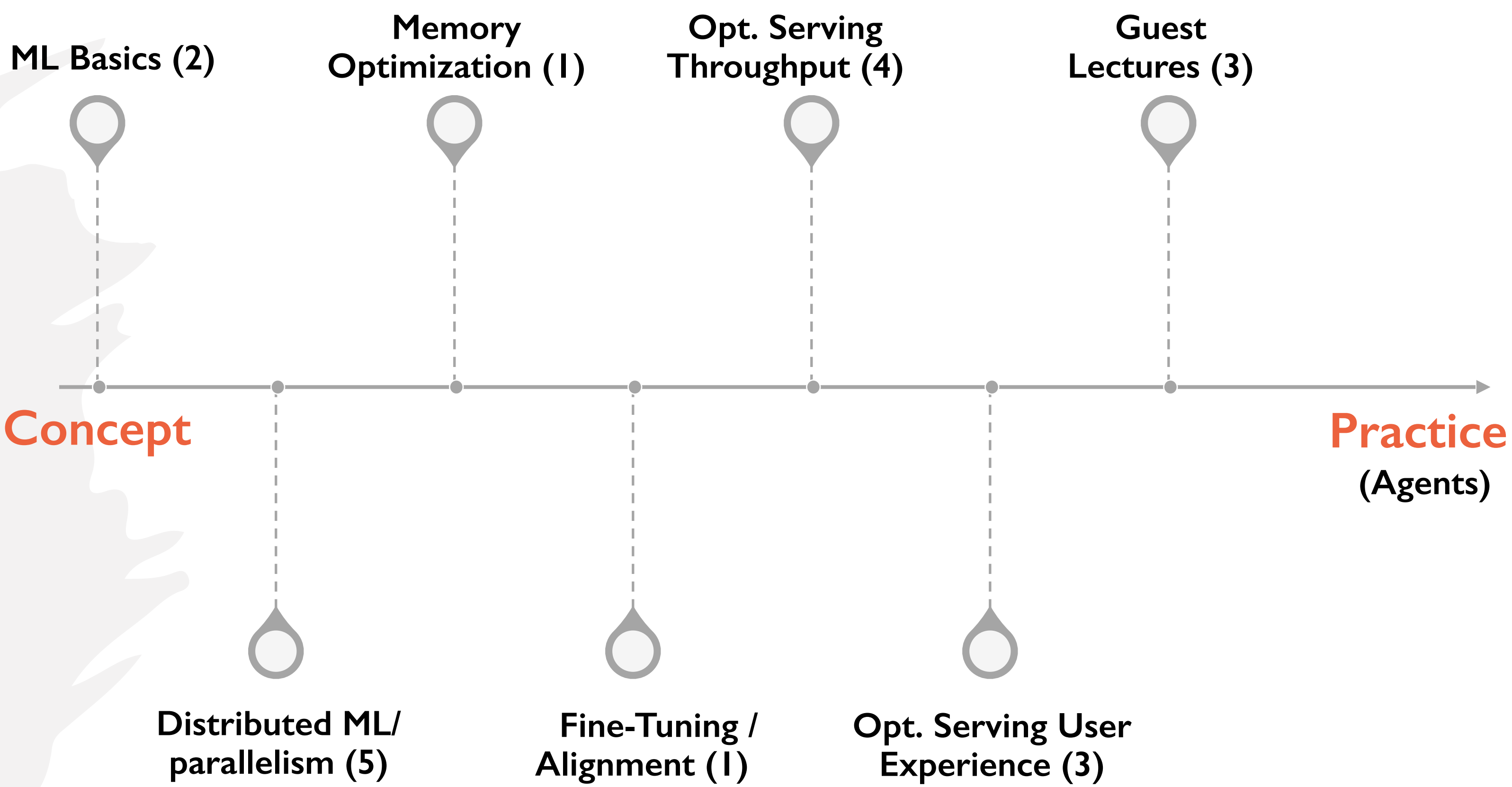
What will you learn in this course?

-- How to Support Practical ML

System for X (#Lectures)



What will you learn in this course?

-- How to Support Practical ML

Learning Outcomes

After completing this course you should be able to

- Understand the basic functioning of ML Sys
- Articulate and critique latest GenAI Sys
- Be prepared to perform new research

This is a Seminar-like Course.
The more active the better.

Have Fun!

Agenda

- What will you learn in this course?
  - GenAI, systems, networks, ...
- What will you do in this course?
  - Participation, assignment, course project, ...

Course Expectations (Grading)

	Weight	Notes
Participation	15%	Up to two absences
(Group) Panel Discussion	8%	Two discussions (4%+4%)

While we value teamwork, clarify each member's contribution in group report.

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Lab Assignments	16%	2-3 assignments (Three for 4 credits)

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	Weight	Notes
Participation	15%	Up to two absences
(Group) Panel Discussion	8%	Two discussions (4%+4%)
Lab Assignments	16%	2-3 assignments (Three for 4 credits)
Reading Summary	16%	Opt in to 8 (x2%) out of 10 readings
(Group) Project Presentation	15%	One final presentation
(Group) Project Report	30%	Proposal, mid-term, final report

While we value teamwork, clarify each member's contribution in group report.

Grading Scheme (grade is the better of the two)

Grade	Absolute Cutoff (>=)	Relative Bin (>=)
A+	95	Top 5%
>= A-	88	Top 35%
>= B-	75	Top 80%
>= C-	60	Hard to Believe!
<C-		Unbelievable!!

Out of 300+ students, only 5 got <= B in my previous course. 😊

Form Groups ASAP

- Use Canvas to find potential group members
- Group size should be 4-5
  - May allow a few smaller groups if/when students drop out
- Submit at https://forms.gle/2YweEPuscwYWGwsk6
  By Sept 5th the latest, but **right now** is better

Participation (15%) & Paper Summaries (16%)

- **Up to two absences in both attendance and paper summaries**
  - Write summaries for *8 out of 10* papers (feel free to pick 8; 2% each)

No class

Date	Topic	Lecturer	Slides	Assignment/Summary
Aug 24	Course Introduction and Logistics	Fan Lai		
Aug 26	**No Class** (Project Kickoff & Team Formation)	Fan Lai		
Aug 31	Transformers	Fan Lai		
Sept 2	Transformers Deep Dive	Fan Lai		
Sept 7	**No Class** (Labor Day)			
Sept 9	Distributed Training Overview	Fan Lai		BLASST

First summary due

Please upload your PDF summary to Canvas by 11:59 PM *before* lecture day.

Participation & Paper Summaries

- **Each summary (4-5 paragraphs) must follow the template**
  - What is the problem and why is it important?
  - The main proposed idea(s).
  - A summary of your understanding of different components of the proposed technique, e.g., the purpose of critical design choices.
  - What are the *technical* drawbacks/limitations
  - How will you improve them?

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  - The main proposed idea(s).
  - A summary of your understanding of different components of the proposed technique, e.g., the purpose of critical design choices.
  - What are the *technical* drawbacks/limitations
  - How will you improve them?

Grading criteria, each summary has 10 points in total. For each item here, you get:

- 2: The summary item demonstrates a clear understanding of the paper.
- 1: The summary item misses the point of the paper.
- 0: The summary item is missing.

Lab Assignments (16%)

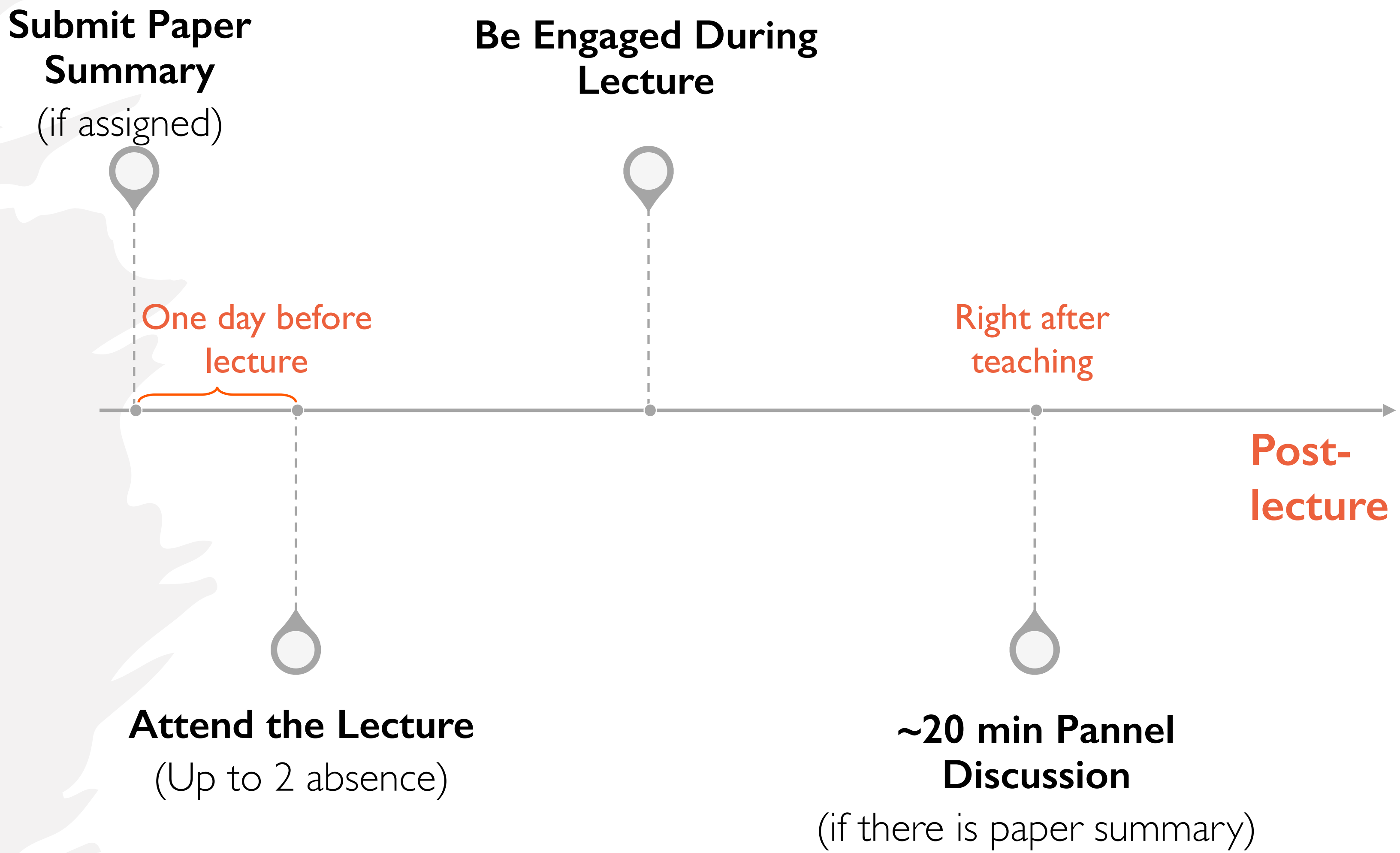
- **Two Assignments for 3-credit students (Three for 4 credits)**
  - Likely will use NCSA clusters for GPUs; Remember to *start early* due to GPU access
  - We will provide details later
- **Topics**
  - Distributed model training
  - Inference optimizations

Panel Discussion (After Normal Lecture)

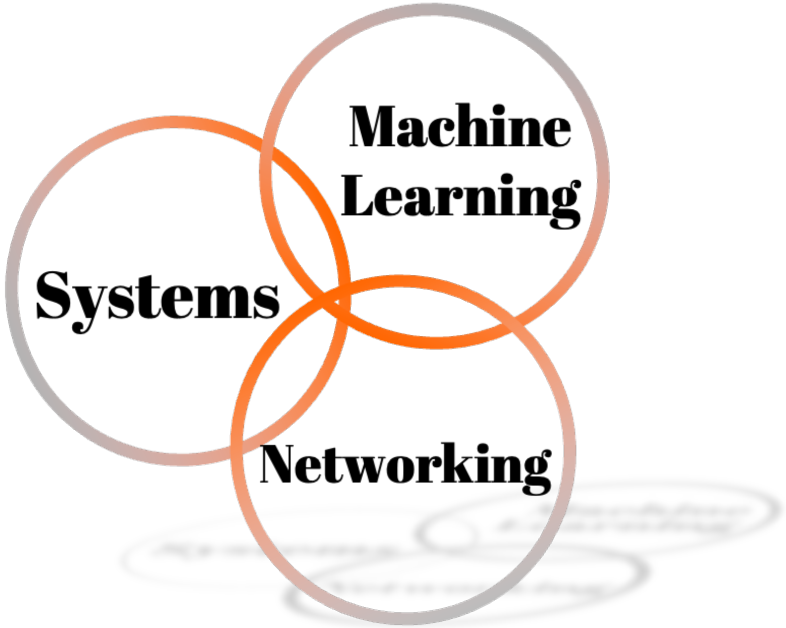
- **The Authors (assigned)**
  - 'Companion' group that answers questions from 'Reviewers'
- **The Reviewers (assigned)**
  - 'Reviewer' group that poses challenging **research** questions
- **Rest of the Class & Instructor**
  - Ask questions directly too

Each group will be assigned to these slots at least once; 2 discussions per group (only! 😊)

Summary of Your Role



Systems for ML Projects



Research-Oriented Course Project

- **The final project accounts for 45% of total grades**

- **What can and cannot be a project?**
  - Just surveys are not allowed
  - Measurements of new environments or of existing solutions in new environments are acceptable
  - Proposing new research problems or addressing existing ones is highly preferred!

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- **What can and cannot be a project?**
  - Just surveys are not allowed
  - Measurements of new environments or of existing solutions in new environments are acceptable
  - Proposing new research problems or addressing existing ones is highly preferred!

- **An ideal project should answer the questions you asked during paper summary and points you cared about in panel discussion**

How to Approach it?

1. **Find a problem and motivate why this is worth solving**
2. **Quickly survey background and related work**
   - Might require you to go back to the first step
3. **Form/update your hypothesis**
4. **Test your hypothesis**
   - Go back to 3 until you are happy
5. **Present your findings to the instructor and in writing**
   - Discuss known limitations

Draft Proposal (Sept 30; 5%)

- Two pages (template), plus references, **must** include
  - *Problem:* What is the problem?
  - *Motivation:* Why is it important?
  - *Solution overview:* Any initial thoughts?
  - *Evaluation plan:* How would you evaluate it?

- Submit as a group, including your group information *(form a group ASAP)*

- Approved by the instructor and agreed upon by you
  - Forms the basis of expectation

Feel free to stop by during this Wednesday's class hours in SC 3128, and we can brainstorm.

Mid-Semester Report (Nov 4; 10%)

- **Dedicated proposal feedback session (No class on Oct 5, 7)**
  - This is to make sure you are receiving feedback and making progress
  - Feel free to stop by *office hours on any day*!

- **Extend your proposal to 4 pages (template), which must include**
  - What is the problem?
  - Why is it important? (Motivating experiments)
  - What are the most related work?
  - What's your hypothesis so far?
  - How are/will you evaluate it?

Final Presentation (Dec 2, 7, 9; 15%)

- **In-class (group) presentation**
  - Please spend a significant amount of time on working your project and making this presentation nice and clear.

- **Graded by instruction team (50%) and your classmates (50%)**
  - **Instructor:** based on format, correctness, depth, clarity, insights
  - **Peers:** make sure your classmates feel they indeed learn something after listening to your presentation

Final Report (Dec 18; 15%)

- **Final report (8 pages excluding references by Dec 18):**
  - Should be written like the papers you've read
    - Abstract, Introduction
    - Background/Motivation
    - Design
    - Implementation and Evaluation
    - Conclusion
  - As if you'd submit it to a workshop with ~3 more months of work or to a conference after ~6 more months of work
  - How to Write a Great Research Paper by Simon Peyton Jones

Project Ideas

- **Some potential project ideas (let's chat more)**
  - Multi-tier memory, including CXL and RDMA, for long-context LLMs
  - Energy-efficient GenAI serving on end-user devices
  - Systems support for reinforcement learning from human feedback
  - Making chatbots real by allowing them to cut in
  - Repurposing spatiotemporal redundancy in video generation
  - Benchmark important DL workloads to identify important angles
  - ML for systems optimizations (e.g., networking problems)
  - …
- **Any of your crazy ideas!**

Feel free to stop by during this Wednesday's class hours in SC 3128, and we can brainstorm.

Thank you!

Date	Milestone	Details
09/05	Form Group	Find 4-5 like-minded teammates
09/30	Submit Proposal	Send your proposal by email to receive feedback either via email or in-person or both
10/05, 10/07	Proposal Feedback (no class)	Refine the problem, overview, and form initial hypothesis/idea
11/04	Mid-term Report	4-page report
12/02 12/07 12/09	In-Class Project Presentations	Present your research findings
12/18	Research paper	Submit a report like the papers you read

NO extensions! Delayed submission will receive NO CREDIT

Thank you!

(No regular class this Wed; Project Kickoff & Team Formation)

Attendance & Summary

- Up to 2 absences
- 8 out of 10 papers

Pannel Discussion

- Companion + Reviewer

Form Groups of 4-5; submit by *Sept 5*

2-3 Lab Assignments

- Distributed training
- Inference optimizations

Semester-long Project

- Final presentation
- Final report